

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

D Speedway: Automotive Repair

A Smart Inventory and Parts Planning System

In Partial Fulfillment of the Requirements
for Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Introduction

D Speedway is a local automotive repair shop that has been serving vehicle owners in the community for a year. The shop handles everything from routine maintenance and oil changes to parts replacement and engine diagnostics. The owner and staff genuinely take pride in the quality of service they provide, and customers keep coming back because of it.

It is the kind of shop where you can walk in not really knowing what is wrong with your car and walk out actually understanding what was fixed and why. That level of transparency and honesty is not something every repair shop offers, and people in the community have picked up on it. Word spreads fast in a neighborhood, and when a shop does right by its customers consistently, the reputation builds on its own. D Speedway has managed to establish that kind of standing in just one year, which is a reflection of how seriously the team takes the work.

The range of services the shop covers also matters. Vehicle owners do not have to go to one place for an oil change and another for a more complex repair — D Speedway handles it all under one roof. That convenience, paired with the trust the shop has built, is a big part of why the customer base has kept growing since the shop opened. More customers means more vehicles coming in, more parts being used, more transactions happening every day. And that growth, as

good as it is for the business, is also what is starting to put pressure on the way the shop manages things behind the scenes.

The problem, though, is not about the quality of their work — it is about how they keep track of everything behind the scenes. Right now, all of their sales records and inventory are managed through a collection of Excel files. On the surface this might sound manageable, but in practice it creates real headaches every single day. When a part is sold, someone has to remember to update the spreadsheet. When two staff members use the same file, entries sometimes conflict. When the owner wants to know how much revenue came in last month, it takes hours of manually sorting through rows of data to find the answer.

The thing about spreadsheets is that they only work as well as the person maintaining them. They do not update on their own. They do not catch mistakes. They do not prevent two people from typing into the same cell at the same time and ending up with corrupted data. Every single entry depends on a person remembering to do it, doing it correctly, and doing it at the right time. In a shop environment where the main priority is fixing cars and serving customers, that kind of consistent, precise administrative work is always going to be the first thing that slips. Not because the staff is irresponsible, but because it is genuinely hard to maintain that level of discipline in the middle of a busy workday.

There is also the issue of how many files are actually involved. It is not just one spreadsheet that tracks everything — it is a collection of them. Parts inventory might be in one file, sales records in another, supplier information somewhere else. When those files are maintained separately and

updated by different people at different times, they start to drift apart. The numbers in one file stop matching the numbers in another. Figuring out the actual state of things requires opening multiple files, cross-referencing them, and trying to reconcile discrepancies that should never have existed in the first place. For a shop that is supposed to be focused on repairs, spending time doing that kind of detective work on spreadsheet data is a real problem.

What makes this harder is that the automotive parts market does not sit still. The demand for certain parts spikes at certain times of the year. Supplier prices change. New vehicle models come out and shift what customers need. A business that can only look at its data after the fact — rather than in real time — will always be one step behind. For a shop like D Speedway that depends on having the right parts ready when mechanics need them, being one step behind is a real cost.

Think about what it means practically when demand for a certain part spikes and the shop does not see it coming. A customer brings in a vehicle that needs a part the shop usually has in stock. But because demand has been higher than usual over the past few weeks and nobody noticed in time to reorder, the part is gone. The customer has to wait. The shop has to reach out to a supplier and try to get it as fast as possible — often paying more for the urgency. The mechanic who was supposed to be working on that vehicle has to move on to something else or sit idle. All of that disruption comes from one thing: not having current, reliable data that could have flagged the trend before it became a problem.

Supplier prices changing is another dimension of this. When prices go up, the shop needs to know quickly so it can adjust what it charges customers or decide whether to stock up before the increase hits. When prices drop, that is an opportunity to order more and improve the margin. But if the person making those decisions is working from spreadsheet data that is days or weeks out of date, those windows close before anyone even realizes they were open. The shop ends up making decisions based on old information, which is only slightly better than making no decision at all.

This project will develop a Smart Inventory and Parts Planning System to address these issues directly. Rather than replacing the way D Speedway operates, the system will give the shop the tools to operate the way they already do — just faster, more accurately, and with much less manual effort.

That distinction is worth holding onto. A lot of software solutions for small businesses come with the assumption that the business needs to change how it operates in order to fit the software. New workflows, new processes, new ways of doing things that the staff has to learn from scratch. That approach tends to create more problems than it solves, especially in a shop environment where people are already good at their jobs and do not have a lot of time to spare for learning a completely new system. The goal here is the opposite — to build something that fits around the way the team already works, not something that asks them to work differently.

What that looks like in practice is a system that runs in the background, handling the data side of things automatically so that the staff can stay focused on what they are actually there to do.

When a part is sold, the inventory count updates on its own. When stock drops below a certain level, an alert goes out so a reorder can be placed before the shelves run empty. When the owner wants to see how the business is doing, the numbers are already there — organized, current, and ready to look at without any manual compiling required. The technology does the administrative work. The team does the automotive work. That is the division of labor this system is designed to create.

To understand why this matters, it helps to think about what a typical day looks like at the shop. A customer brings in a vehicle. A mechanic checks it and identifies the parts needed. Someone then has to find out whether those parts are in stock — which means either going to check the storage area physically or opening up one of the Excel files and hoping the number there is still accurate. If the part is available, great. If it is not, the customer has to wait, and the shop has to scramble. That kind of situation happens more often than it should, and it all traces back to not having a reliable, up-to-date record of what is on hand.

That moment — the one where someone has to stop what they are doing and go find out if a part is actually available — is where a lot of time quietly gets lost. It does not seem like much on its own. A few minutes here, a short walk there. But when it happens multiple times a day, across multiple staff members, it starts to add up in a way that is hard to ignore. The shop is paying for that time, even if nobody has ever sat down and calculated what it costs. And beyond the time, there is the uncertainty of it. Even after checking, there is no guarantee that the number in the spreadsheet or the count on the shelf matches reality, because both depend on records that may not be fully up to date.

When a customer has to wait because a part is not in stock, the effect goes beyond just that one visit. It changes the experience they have with the shop. Most people are understanding the first time it happens, but if it becomes a pattern — if they come in and are told to wait more than once because the shop did not have what it needed — they start to wonder whether they should be going somewhere else. Customer loyalty is not unconditional. It is built up through positive experiences and worn down by frustrating ones. D Speedway has worked hard to earn the loyalty it has, and it would be a shame to see that eroded by an inventory problem that is entirely solvable.

There is also the scramble itself to consider — the frantic calls to suppliers, the rush orders, the reshuffling of work to cover for a vehicle that cannot be finished yet. That kind of reactive mode is stressful for the staff and expensive for the business. Rush orders almost always cost more than planned orders. The time spent sorting out the situation is time that could have gone toward other jobs. And the whole thing could have been avoided if the shop had known, even a day or two earlier, that the part was running low and needed to be reordered.

It is also worth pointing out that D Speedway has only been open for a year. The problems they are experiencing now will only get worse as the shop gets busier and the number of parts they carry grows. A system that barely holds together with the current workload is not going to handle a larger one. Addressing this early, while the shop is still relatively small, is much smarter than waiting until the problems become impossible to manage.

A year in business is still early. The habits and systems a business builds in its first couple of years tend to stick around for a long time, for better or worse. If D Speedway keeps running on spreadsheets as it grows, those spreadsheets will become more complicated, more fragile, and more expensive to maintain. More parts to track means more rows, more files, more chances for something to go wrong. More staff means more people touching the same data, which means more conflicts and inconsistencies. The problems that are annoying now become serious problems later, and serious problems are a lot harder to fix than annoying ones.

There is a version of this where the shop waits until things break down badly enough to force a change. That version is more expensive, more disruptive, and more stressful than dealing with it now. The data that has been entered incorrectly over the years would need to be cleaned up. The habits that formed around a broken system would need to be unlearned. The staff would need to be trained on a new system in the middle of a busy operation rather than at a manageable pace. None of that is impossible, but all of it is harder than just building the right foundation now, while the shop is still at a scale where change is straightforward.

The idea behind this project is straightforward. Instead of relying on staff to remember to update a spreadsheet after every transaction, the system handles that automatically. Instead of the owner having to manually compile data from different files to get a clear picture of the business, the system presents that information whenever it is needed. The shop keeps doing what it does — the system just takes care of the record keeping in the background.

What this project is really about, underneath all the technical details, is giving D Speedway a fair shot at growing into what it has the potential to become. The shop already has the most important thing — a team that cares about doing good work and a customer base that trusts them because of it. That foundation is solid. What it needs now is an operational layer that matches the quality of what happens on the shop floor. A system that is as reliable as the mechanics, as consistent as the service, and built to hold up as the business keeps moving forward.

A Smart Inventory and Parts Planning System will not change the soul of D Speedway. The owner will still care about every job. The mechanics will still take pride in their work. The customers will still come back because they trust the shop. But the stress of not knowing what is in stock, the hours lost to manual data entry, the scramble when something runs out unexpectedly, the frustration of not being able to get a clear read on the business without spending half a day on it — all of that goes away. And what is left is a shop that is not just good at what it does, but well-equipped to keep doing it for a long time.

1.2 Statement of the Problem

D Speedway is a local automotive repair shop that has been serving its community for one year. In that short span of time, the shop has managed to build something that most businesses spend years working toward — a genuinely loyal customer base and a reputation that speaks for itself. The team handles everything from routine oil changes and preventive maintenance to full parts replacements and complex engine diagnostics. Customers return not just because the work is done well, but because the staff takes the time to explain what was wrong, what was done, and

why. That kind of transparency is rare in the automotive industry, and the community has noticed.

The shop's growth over its first year has been steady and organic, driven almost entirely by word of mouth. More customers have meant more vehicles coming through the doors, more parts being used, and more transactions happening every single day. That growth is a sign that the business is doing the right things on the shop floor. The problem, however, is that the systems running behind the scenes have not grown alongside it. While the quality of service has remained consistently high, the administrative and operational infrastructure supporting the business is beginning to show serious cracks — and those cracks are becoming more difficult to ignore with each passing month.

The shop currently manages all sales records and inventory through a collection of manually maintained Excel spreadsheets. There is no unified platform, no automated tracking, and no centralized system that connects what happens at the counter to what is on the shelves. Every update depends on a staff member remembering to make it. Every report requires someone to manually pull together data from multiple files. Every stock check involves either opening a spreadsheet that may or may not be current or physically walking to the storage area to count parts by hand. For a business that is growing as quickly as D Speedway, this approach is no longer sustainable.

The problems identified in this section were not derived from assumptions or general knowledge about how small businesses typically operate. They were discovered through direct engagement with the people at D Speedway. The research team visited the shop, spent time observing how

day-to-day operations are carried out, and held conversations with both the owner and the staff members who deal with these issues on a regular basis. The picture that emerged from those visits was consistent and specific: three interconnected operational problems are actively affecting the shop's efficiency, the reliability of its records, and the quality of the experience it is able to provide to its customers. Each of those problems is described in full detail below.

- **Inefficient Sales and Inventory Tracking** — The most immediately visible problem at D Speedway is the absence of any unified system for recording sales and monitoring inventory in an integrated way. The shop currently relies on several separate Excel files, each one maintained independently and serving a different purpose. Parts inventory lives in one file. Sales records live in another. Supplier details and purchase history may be scattered across additional documents. None of these files are automatically linked to each other, which means that a transaction recorded in the sales file does not automatically update the inventory file — that step has to be done manually, separately, by a person who is likely in the middle of handling other responsibilities at the same time.

This fragmented structure makes even routine reporting unnecessarily complicated. If the owner needs to know how many units of a specific part were sold over the past month, answering that question requires opening multiple files, manually extracting the relevant figures, combining them in a way that accounts for any inconsistencies between files, and then verifying the result against other records to make sure nothing was double-counted or missed. Staff members reported that this kind of exercise can take up an entire afternoon — not because the question is

complicated, but because the system makes it complicated. A business should be able to answer basic questions about its own performance in seconds, not hours.

Beyond the time cost, this setup also means there is no single, authoritative source of truth for the business. When two files contain slightly different numbers for the same item, there is no reliable way to determine which one is correct without doing additional investigation. The owner cannot look at one screen and get an accurate picture of where the business stands. Every decision — about purchasing, about staffing, about which services to prioritize — has to be made without the benefit of clean, consolidated, readily accessible data. That is a significant handicap for any business, but it is an especially costly one for a shop that is growing quickly and needs to be making those decisions more frequently and more accurately as it scales.

- **Inaccurate and Unreliable Records** — The second problem is a direct consequence of the first, and in many ways it is more serious. Because every entry in the shop's records depends on a person manually updating a spreadsheet — at the right time, in the right file, with the right information — the records are only as reliable as the conditions under which those entries are made. In a busy shop environment where the primary focus is on fixing vehicles and serving customers, maintaining that level of consistent, precise administrative discipline is genuinely difficult. Not because the staff is negligent, but because the environment itself works against it.

The types of errors that result from this setup are varied but consistent. A part gets used on a repair, but the inventory count does not get updated until the end of the day — or sometimes not at all. A sale is recorded in the sales file but the corresponding entry in the inventory file is

skipped because two things needed attention at the same time. A part gets counted in two separate entries because it was logged once during receiving and again during a stock check. These mistakes are individually small, but they accumulate over time. After weeks and months of small errors layering on top of each other, the figures in the spreadsheets no longer reflect what is actually on the shelves.

What happens next is the part that makes this problem especially damaging. Once the staff realizes that the records cannot be fully trusted, they stop using them as their primary reference. Instead of consulting the spreadsheet to check whether a part is in stock, a staff member walks to the storage area and counts it manually. Instead of relying on a sales report to assess recent performance, the owner makes decisions based on memory and general impression. The records — which exist precisely to reduce that kind of guesswork — become something the team works around rather than with. Any report or purchasing decision that is generated from those files carries an unacknowledged margin of error, and the people making decisions based on those reports may not always know how wide that margin is. That uncertainty is a problem that compounds quietly, and it affects the quality of every business decision downstream.

- **Absence of Real-Time Inventory Visibility and Proactive Stock Alerts** — The third problem is the one that customers feel most directly, even if they never understand the reason behind it. D Speedway currently has no system for monitoring inventory levels in real time and no mechanism for automatically alerting staff when a part is running low. The only way to know the current status of any item in stock is to either consult a spreadsheet that may be out of date or physically go to the storage area and look. Neither approach is reliable, and neither supports the kind of forward-looking planning that a growing business needs.

The consequences of this gap have already played out on the shop floor in ways that directly affect both operations and customer experience. Mechanics have begun repairs only to discover partway through the job that the part they need is no longer available. At that point, the entire job has to stop. The customer, who may have dropped off their vehicle expecting it to be ready within a set timeframe, now has to wait while the shop contacts a supplier to source the missing part — often on short notice, which typically means paying a higher price for expedited delivery. Meanwhile, the mechanic either sits idle or shifts to another vehicle, disrupting the shop's workflow and delaying other customers in the process.

The frustration this creates for customers is real and cumulative. A single instance of unexpected delay is something most people can accept. But when it happens more than once, or when a customer hears about it happening to someone they know, it starts to affect how they perceive the shop. Customer loyalty is not unconditional — it is built through positive experiences and worn down by negative ones. D Speedway has invested a great deal in building the trust it currently has, and avoidable inventory shortages represent a recurring risk to that investment. The missed opportunity cost is also worth noting: every job that gets paused due to a stock shortage is a job that is not generating revenue, and every emergency reorder paid at a premium is money that planned purchasing would have saved.

What makes this problem particularly worth emphasizing is how straightforwardly it could be prevented. An automated alert system that notifies staff when a part's stock level falls below a defined threshold would give the shop days of advance warning before a shortage becomes a disruption. A reorder could be placed during normal purchasing cycles, at standard pricing, with

enough lead time to ensure the part arrives before it is needed. The repair would proceed without interruption. The customer would leave on time. The mechanic would stay productive. None of that requires complex technology — it just requires a system that is actually designed to support the way the shop operates, rather than one that leaves critical information buried in files that no one is regularly monitoring.

How These Problems Are Connected

Understanding each of these problems individually is important, but understanding how they relate to each other is equally critical — because the relationship between them explains why a partial solution would not be enough to meaningfully improve the situation at D Speedway. These are not three separate issues that happen to coexist in the same business. They are three stages of a single, compounding failure that begins with how data is entered, affects how reliable that data becomes over time, and ultimately determines how much useful information the business can extract from it.

The tracking problem is the origin point. When sales and inventory are managed through a collection of manually updated, disconnected spreadsheets, the conditions for error are present at every transaction. There is no automation catching missed entries. There is no validation checking whether a recorded sale is consistent with current stock levels. There is no flag raised when two staff members enter conflicting information about the same item. Every update is a

manual act, and every manual act is an opportunity for something to go wrong. Over time, those opportunities are taken — not out of negligence, but simply because the volume of work and the pace of a busy shop make perfect administrative consistency an unrealistic expectation.

The errors that accumulate from the tracking problem feed directly into the accuracy problem. Small discrepancies compound into larger ones. Files that were once reasonably current begin to drift further and further from reality. Staff members who once trusted the records begin to doubt them, and eventually stop using them as a reliable reference. The system designed to reduce guesswork ends up introducing more of it. Reports generated from those files carry errors that may not be obvious to the person reading them. Purchasing decisions made on the basis of those reports may be based on stock levels that do not reflect what is actually on the shelf. The entire administrative layer of the business becomes unreliable, and that unreliability has downstream consequences for every decision the owner makes.

The visibility problem then emerges from the accuracy problem in a way that closes the loop. Even if the shop wanted to implement some form of real-time stock monitoring, the data feeding that monitoring would not be trustworthy enough to act on confidently. An alert generated from inaccurate records might flag a shortage that does not actually exist, or — more dangerously — fail to flag one that does. The value of visibility depends entirely on the reliability of the data behind it, and as long as the underlying records are compromised by manual entry errors and disconnected files, no amount of monitoring infrastructure will produce results that can be genuinely trusted.

This chain of dependency is the reason this project addresses all three problems together rather than in isolation. Fixing the tracking problem by centralizing and automating data entry creates the conditions for accurate records. Accurate records make real-time monitoring meaningful rather than misleading. Meaningful monitoring gives the owner and staff the visibility they need to plan ahead, respond proactively to stock changes, and make purchasing decisions based on current, verified information rather than outdated figures and educated guesses. Each component of the solution reinforces the others, and together they address the full scope of what D Speedway is dealing with in a way that no single fix could accomplish on its own.

Why This Needs to Be Addressed Now

One detail worth keeping in mind throughout all of this is that D Speedway has only been open for one year. The problems described above are already present and already causing disruption at the shop's current scale. That fact alone should prompt urgency — because if these problems are visible now, they will become significantly worse as the business continues to grow. More customers means more transactions. More transactions means more data to track. More data to track means more opportunities for the manual system to fail, and more time required to maintain it even poorly. A system that is already struggling under the current workload is not equipped to handle a heavier one.

There is also the question of embedded habits. The systems a business relies on during its formative years tend to become deeply ingrained — not just in the tools and processes, but in the daily routines and mental models of the people who use them. If D Speedway continues building

on its current spreadsheet-based foundation as it grows, those spreadsheets will become more elaborate, more difficult to maintain, and more deeply woven into how the team operates. At some point, the cost of replacing them — in terms of data migration, staff retraining, and operational disruption — becomes significantly higher than it would be today. The longer the shop waits, the harder the transition becomes.

The alternative scenario — where nothing changes until the problems become severe enough to force a crisis — is substantially more costly and disruptive than acting now. Data that has been entered incorrectly over months or years would need to be audited and corrected. Workflows that developed around a broken system would need to be redesigned from the ground up. Staff would need to be trained on a new system in the middle of normal operations, without the breathing room that early adoption provides. That is a difficult and expensive process under any circumstances, and it is made considerably more difficult when the business is already under the pressure of day-to-day demand.

Addressing these problems now, while D Speedway is still in its early growth phase and while the scale of change remains manageable, is both the practical and the strategic choice. The shop already has the most important things in place: a team that takes pride in its work, a customer base that trusts it, and a reputation built on genuine quality and transparency. What it needs is an operational foundation that is as solid and reliable as the service it provides. A Smart Inventory and Parts Planning System will not change what makes D Speedway worth coming back to. It will simply ensure that the business has the tools it needs to sustain and build on everything it has already worked to establish.

1.3 Objectives of the Study

1.3.1 General Objective

The goal of this project is to design and develop a Smart Inventory and Parts Planning System for D Speedway that will bring order to how the shop manages its sales data and stock levels. The system will make it easier for staff to record transactions, give the owner real-time visibility into what is happening in the business, and take the guesswork out of deciding when to reorder parts.

This goal was shaped by the actual needs of the shop, not just general best practices. The features being built are a direct response to the problems described in Section 1.2. Every decision made during development — what to include, how it should work, how it should look — goes back to the question of whether it makes things easier for the people at D Speedway.

It is also important that the system be something the staff can actually use without a lot of training. Not everyone at the shop is comfortable with technology, and a system that requires extensive technical knowledge to operate would not help anyone. The design needs to be simple and clear so that even someone using it for the first time can figure out what to do without needing step-by-step instructions.

For the owner, the main benefit is being able to see what is happening in the business at any given moment. Right now, getting that picture requires a lot of manual effort. With this system,

the information is just there whenever it is needed — current stock levels, recent sales, which parts are running low. That kind of visibility makes it much easier to run the business well.

1.3.2 Specific Objectives

To achieve this, the study will:

- Build a module for recording and tracking sales on a daily, monthly, and yearly basis — replacing the current multi-file Excel setup with a single, centralized tool.
- Implement inventory monitoring that updates stock levels automatically after every transaction and sends alerts when parts fall below a set minimum.
- Integrate a forecasting feature that looks at past sales patterns to predict which parts the shop will likely need in the weeks ahead.
- Create a reorder recommendation tool that suggests how many units of a part to order and flags which supplier to use based on previous purchases.
- Test the finished system with D Speedway staff and evaluate it against criteria for usability, accuracy, and performance.

The first objective — the sales tracking module — serves as the foundation of the entire system. Everything else in the project depends on whether sales data is being recorded accurately and consistently. Right now, D Speedway relies on several Excel files that are updated separately, which makes tracking sales more difficult than it should be. Some records are stored in one file

while other information is placed somewhere else, and because updates are done manually, there is always the possibility that something gets missed, entered incorrectly, or updated late.

The purpose of creating a centralized sales tracking module is to simplify that process and organize all transaction records into one system. Instead of staff needing to move between multiple spreadsheets just to record a sale or review past transactions, the system will store everything in a single platform that can be accessed more easily. Daily, monthly, and yearly sales reports will also be generated automatically, which means the owner no longer has to spend hours manually sorting and compiling data just to understand how the business is performing.

This objective is especially important because sales data affects nearly every other part of the system. If transactions are not being recorded correctly, then inventory counts become inaccurate, reports lose reliability, and forecasting results become less useful. In other words, the system cannot make good inventory decisions if the sales information feeding into it is incomplete or incorrect. That is why getting this module right is considered one of the most important parts of the entire project.

Another important benefit of this module is visibility. Once sales records are centralized, the owner will be able to identify which automotive parts are selling frequently, which services are most common, and which periods of the month or year bring in the highest amount of business activity. Having access to this kind of organized information makes decision-making much easier because the owner no longer has to rely only on memory or rough estimates.

The second objective, automatic inventory monitoring, focuses on solving one of the shop's biggest operational problems: the lack of reliable and updated stock information. At present,

inventory tracking depends heavily on manual updates and physical checking of parts. In a busy repair shop environment where mechanics and staff are focused mainly on repairs and customer service, updating spreadsheets accurately after every transaction is difficult to maintain consistently. As a result, stock counts often become unreliable over time.

The proposed system aims to remove much of that manual work by automatically updating inventory records whenever a transaction is completed. Every time a part is sold or used during a repair, the system immediately adjusts the stock count without requiring additional encoding from the staff. This creates a more accurate and real-time view of inventory levels, reducing the chances of discrepancies between recorded stock and actual physical inventory.

This objective is important not only for improving record accuracy but also for improving the overall workflow inside the shop. Mechanics and staff members will no longer need to stop what they are doing just to physically confirm whether a part is available. Instead, they can quickly check the system and view updated stock information immediately. This saves time, reduces interruptions, and allows the shop to operate more efficiently throughout the day.

Another major feature included in this objective is the low-stock alert system. The system will notify staff whenever the quantity of a part reaches a minimum level set by the shop. This helps prevent situations where parts unexpectedly run out during repairs. Instead of discovering shortages only after a customer's vehicle is already being serviced, the shop can identify low inventory levels earlier and reorder parts before they become unavailable.

This proactive approach is much more practical than the current reactive process. Right now, inventory shortages often create unnecessary stress because staff members only realize a part is unavailable when it is already urgently needed. By providing early warnings, the system allows the shop to plan ahead, avoid delays, and maintain smoother operations.

The third objective involves integrating a forecasting feature into the system. This part of the project is intended to help D Speedway prepare for future inventory needs by analyzing previous sales and inventory movement patterns. Rather than depending entirely on guesswork or personal judgment, the system will use historical data to estimate which automotive parts are likely to be needed more frequently in the coming weeks or months.

Forecasting is important because inventory demand in automotive repair shops is rarely completely stable. Certain parts may become more in demand depending on the season, the type of vehicles commonly serviced, or recurring repair trends within the community. Without a reliable way to identify these patterns, businesses often end up reacting to shortages after they already happen instead of preparing ahead of time.

The forecasting feature gives the shop a more organized way of planning inventory purchases. By reviewing historical sales records, the system can identify which items move quickly, which parts are used consistently, and which products may need restocking soon. This allows the owner to make purchasing decisions based on actual trends instead of assumptions alone.

Although forecasting cannot guarantee perfect predictions, it still provides a significant advantage compared to operating without any planning support at all. Even a reasonable estimate based on past sales patterns can help reduce stock shortages, improve inventory preparation, and make purchasing more efficient. Instead of constantly operating in a reactive mode, the shop can begin making more proactive decisions regarding inventory management.

The forecasting feature also supports better financial planning. Overstocking unnecessary parts ties up money that could have been used elsewhere in the business, while understocking important items causes delays and missed opportunities. By helping the shop better understand inventory movement, forecasting contributes not only to operational efficiency but also to more practical use of business resources.

The fourth objective focuses on creating a reorder recommendation tool that simplifies the restocking process. While low-stock alerts inform staff that inventory levels are becoming critical, the reorder recommendation feature takes things one step further by helping determine what actions should be taken next.

When the system detects that a part is running low, it will provide suggested reorder quantities based on previous purchasing history and inventory movement. This reduces the need for staff to manually review old records just to estimate how much stock should be purchased. Instead of relying entirely on memory or rough calculations, the owner can make decisions using organized and updated information generated by the system.

The system will also identify which supplier has previously been used for certain parts whenever supplier information is available in the database. This feature helps make the purchasing process more organized and efficient because staff no longer need to search through different files or receipts just to identify where a part was previously ordered from.

This objective is especially useful for reducing delays during restocking. In many small businesses, inventory shortages create stressful situations where employees must quickly contact suppliers, compare prices, and decide how much inventory to purchase within a short amount of time. Having a system that already provides reorder suggestions simplifies that process and helps the shop respond faster whenever inventory replenishment becomes necessary.

The reorder recommendation tool also supports better inventory control overall. Instead of ordering parts randomly or based purely on estimation, the shop can make more consistent purchasing decisions using actual inventory movement data. This helps minimize both overstocking and understocking problems while improving the overall organization of inventory planning.

Finally, the fifth objective focuses on testing and evaluating the completed system using actual staff members from D Speedway. This stage is just as important as the development process

itself because a system cannot be considered successful if it works technically but is difficult or confusing for people to use in real working conditions.

The testing process will evaluate the system based on usability, accuracy, functionality, and overall performance. Staff members will be asked to use the system during normal operational tasks so the researchers can observe whether the features are understandable, responsive, and practical within the actual shop environment. Their feedback will help determine whether the system effectively addresses the operational problems identified in the study.

Usability is particularly important because not all employees are highly experienced with technology or computerized systems. The system needs to remain simple enough that staff members can use it confidently without requiring extensive technical training. A complicated system may create additional frustration instead of solving existing problems, which is why user experience is considered a major part of the evaluation process.

Accuracy testing will also be conducted to ensure that inventory updates, stock alerts, forecasting outputs, and sales reports are functioning correctly. Since the system is intended to replace manual spreadsheet-based tracking, the reliability of the information it generates is critical. The shop needs to be able to trust the records produced by the system in order for it to become a practical tool for daily operations.

Performance evaluation will further examine how efficiently the system handles transactions, retrieves records, and processes inventory updates. Since automotive repair shops operate in fast-

paced environments, the system must remain responsive and capable of handling daily operational demands without causing delays.

Ultimately, the testing phase ensures that the proposed Smart Inventory and Parts Planning System is not only technically functional but also realistic, reliable, and suitable for the actual needs of D Speedway. The goal of the project is not simply to create software, but to develop a practical system that genuinely improves how the shop manages inventory, sales, and operational planning in everyday situations.

1.4 Scope and Delimitation

This study focuses on the design and development of a Smart Inventory and Parts Planning System for D Speedway, a local automotive repair shop that currently manages its sales and inventory records through manually maintained Excel spreadsheets. The project is intended to address the operational difficulties the shop experiences in tracking sales, monitoring stock levels, planning future inventory needs, and organizing purchasing decisions. The system is being developed specifically around the actual workflow and daily operations observed at D Speedway, with the goal of making inventory management more accurate, organized, and easier to handle without adding unnecessary complexity to the staff's work.

The study covers the development of a centralized system that combines sales recording, inventory monitoring, forecasting, and reorder planning into a single platform. Instead of maintaining separate files for different records, the system will store and manage information in

one connected database so that updates made in one part of the system are automatically reflected throughout the rest of it. This removes the need for repetitive manual updating and reduces the risk of inconsistencies between records.

One of the main areas covered by the study is sales tracking. The system will allow staff to record daily transactions involving automotive parts and services performed at the shop. These records will then be organized automatically into daily, monthly, and yearly summaries that the owner can access whenever needed. The purpose of this feature is to eliminate the current process of manually reviewing and compiling spreadsheet data just to generate basic business reports. Through the system, the owner will be able to monitor recent sales activity, identify frequently purchased parts, and review the overall performance of the business in a more efficient way.

The study also includes inventory monitoring and stock management. Every time a part is sold or used during a repair, the system will automatically update the inventory count in real time. This feature is designed to reduce the reliance on physical stock checking and prevent situations where the records no longer match the actual quantity of parts on hand. The system will also include low-stock notifications that alert staff whenever a part reaches a predefined minimum quantity. These alerts are meant to give the shop enough time to reorder needed parts before they run out completely.

Another major scope of the study is the forecasting and reorder planning feature. The system will analyze previous sales data to identify usage patterns for automotive parts commonly used by the

shop. Based on these patterns, the system will generate simple forecasts that estimate which parts may be needed in the coming weeks or months. The forecasting process is not intended to provide perfect predictions, but rather to help the owner make more informed purchasing decisions using available historical data instead of relying entirely on memory or estimation.

Alongside forecasting, the study also covers reorder recommendations. When a part reaches a low-stock level, the system will not only issue an alert but will also provide suggested reorder quantities based on previous purchasing activity and stock movement. Supplier information previously entered into the system can also be referenced during this process to help the owner identify which supplier has been used for certain parts in the past. This feature is intended to simplify the restocking process and reduce the time spent manually reviewing records before placing orders.

The study further includes the testing and evaluation of the developed system using criteria such as usability, accuracy, functionality, and performance. Actual staff members from D Speedway will participate in the testing process to determine whether the system is understandable, practical, and suitable for real shop operations. Feedback gathered during testing will help identify areas that may need adjustments or improvements before the final implementation of the project.

Although the study covers several important operational areas, there are also clear limitations to what the system is designed to do. The project is focused only on inventory management, sales recording, parts forecasting, and reorder planning within D Speedway. It does not cover every possible aspect of running an automotive repair business.

The system will not include advanced accounting functions such as payroll processing, tax computation, full financial auditing, or complete bookkeeping services. While the system may generate sales summaries and inventory-related reports, it is not intended to replace professional accounting software or the services of an accountant. Financial management beyond sales and inventory monitoring remains outside the scope of the study.

The project is also limited to the operations of a single repair shop branch. The system is designed specifically for the workflow, size, and operational structure of D Speedway as observed during the researchers' visits and interviews. It is not intended to function as a large-scale enterprise system capable of managing multiple branches, warehouses, or nationwide supplier networks.

In addition, the forecasting feature of the system is based only on historical sales records entered into the database. The system will not analyze external market factors such as economic conditions, supplier shortages, seasonal industry disruptions, or changes in automotive trends outside the available data. As a result, forecasting results should be understood as decision-support recommendations rather than guaranteed predictions.

The study is likewise limited by the accuracy of the data entered into the system. Although automation reduces many of the problems associated with manual spreadsheets, the system still depends on staff correctly recording transactions and inventory information. If inaccurate or incomplete data is entered, the reports, forecasts, and stock monitoring generated by the system

may also become inaccurate. The system improves record management, but it cannot completely eliminate human error.

The project also does not include online customer booking, mobile application development, online payment processing, or direct integration with supplier ordering platforms. The focus of the study is strictly on improving the shop's internal inventory and sales management operations rather than developing a full customer management or e-commerce system.

Another limitation of the study is the timeframe available for development and testing. The evaluation of the system will be conducted within the period allowed for the completion of the research project. Because of this, long-term operational testing across several years of business activity will not be included. The study will instead evaluate whether the system performs properly under the conditions and sample data available during the testing phase.

Despite these limitations, the researchers believe that the proposed Smart Inventory and Parts Planning System can still provide significant improvements to the way D Speedway handles its daily operations. The study is centered on solving the specific problems currently affecting the shop, particularly the inefficiency of spreadsheet-based tracking, the unreliability of manually updated records, and the absence of real-time inventory visibility. By focusing directly on these operational concerns, the project aims to provide D Speedway with a more organized, dependable, and practical foundation that can support the continued growth of the business in the years ahead.

1.5 Significance of the Study

The people and groups who stand to benefit from this project include:

- D Speedway (Business Owner and Management):

The business owner and management of D Speedway Auto Repair Shop will be the primary beneficiaries of the proposed Smart Inventory and Parts Planning System. At present, the shop relies heavily on manually maintained Excel spreadsheets to track sales, inventory, and stock movement. While this method may still function for basic monitoring, it becomes increasingly difficult to manage accurately as daily transactions continue to grow.

Because inventory updates depend entirely on manual encoding, records can easily become outdated, inconsistent, or incomplete, especially during busy operational hours when staff members are focused on customer service and vehicle repairs.

Through the implementation of the proposed system, the owner will gain access to a centralized platform capable of recording sales, monitoring inventory, and generating reports in real time. Instead of spending hours manually checking spreadsheets and verifying stock counts, management will be able to view updated information immediately through the system. This provides a clearer and more reliable picture of the shop's actual operations, including which parts are selling quickly, which products remain unused for long periods, and which suppliers consistently provide inventory on time.

The proposed system also helps improve decision-making within the business. Since inventory records and sales data will be stored accurately and consistently, the owner can make better purchasing decisions based on actual inventory movement rather than assumptions or incomplete information. This reduces the risk of unnecessary overstocking while also minimizing situations where important automotive parts suddenly become unavailable during repairs.

In the long term, the value of having organized and reliable business data extends beyond solving current inventory problems. As D Speedway continues to grow, the system can serve as a foundation for improving operational planning, budgeting, forecasting, and business expansion. Accurate records make it easier to identify business trends, monitor seasonal demand patterns, and evaluate operational performance over time. At present, much of this information exists only in scattered spreadsheets that are difficult to organize and analyze effectively. The proposed system transforms that raw data into useful information that management can use to support future business growth and operational improvement.

Another important benefit for management is the reduction of administrative workload. Repetitive tasks such as manually updating inventory files, cross-checking records, and generating reports consume a significant amount of time that could otherwise be spent managing the business itself. By automating these processes, the system allows the owner to focus more on improving services, handling customers, and expanding business operations rather than constantly correcting inventory inconsistencies.

- **Shop**

Staff:

The employees and staff members of D Speedway will also benefit significantly from the implementation of the proposed system. In the current setup, many operational tasks still depend on manual processes, including recording sales transactions, updating inventory counts, and checking stock availability. These tasks may appear manageable individually, but in a busy automotive repair environment where multiple repairs and transactions happen throughout the day, even small delays and repetitive administrative work gradually become time-consuming and frustrating for employees.

The proposed Smart Inventory and Parts Planning System is designed to reduce these daily operational difficulties by simplifying the inventory management process. Recording transactions will become faster and more organized because staff members will no longer need to switch between multiple spreadsheets or manually update separate files after every sale or repair. Once a transaction is recorded, inventory levels will automatically update within the system, reducing the possibility of forgotten entries or inconsistent stock counts.

This automation minimizes human error and reduces the pressure placed on employees to constantly double-check records manually. Staff will no longer need to spend unnecessary time searching for the correct spreadsheet, confirming whether another employee already updated a file, or physically inspecting storage areas just to verify if a part is available. These small inconveniences may seem minor individually, but together they consume a large portion of the workday and can slow down overall operations.

The system also improves workflow coordination among employees. Since all users will access the same centralized database, inventory information becomes more consistent and easier to monitor. Employees can quickly check stock availability, monitor low-stock items, and identify needed parts without relying on verbal communication or handwritten notes. This creates a more organized and efficient working environment where tasks can be completed faster and with fewer interruptions.

In addition, reducing repetitive administrative work allows staff members to focus more on their actual responsibilities, particularly customer service and vehicle repairs. Mechanics and service personnel can concentrate on completing repairs efficiently rather than constantly worrying about whether inventory records are accurate. This not only improves productivity but also reduces workplace stress caused by disorganized inventory management and missing stock information.

Another important benefit is that the proposed system is designed to remain user-friendly and practical for everyday use. The system is not intended to complicate operations or require advanced technical skills from employees. Instead, it is designed around the actual workflow of the repair shop so that staff members can easily adapt to it with minimal difficulty. This increases the likelihood of successful adoption and consistent use within the business.

● **Vehicle Owners (Customers):**

Although customers may never directly interact with the proposed system itself, they are still among the most important beneficiaries of the project because improvements in inventory

management directly affect the quality and speed of service they receive. In automotive repair businesses, customer satisfaction is heavily influenced by how quickly repairs can be completed and whether needed parts are immediately available during servicing.

At present, inventory inaccuracies and delayed updates sometimes create situations where mechanics discover that a required part is unavailable only after a repair has already started. These situations delay repairs, interrupt workflow, and force customers to wait longer than expected while replacement parts are sourced from suppliers. In some cases, customers may even need to return at a later date because the necessary parts are not currently in stock.

By implementing real-time inventory monitoring and automated stock updates, the proposed system helps reduce these disruptions significantly. Mechanics and staff members will be able to verify stock availability immediately before repairs begin, reducing the likelihood of unexpected shortages during service operations. Low-stock alerts and reorder recommendations also help ensure that commonly used automotive parts remain available when needed.

As a result, customers benefit from faster service completion, more accurate repair scheduling, and fewer delays caused by inventory problems. Repairs become more organized because the shop can prepare needed parts in advance rather than reacting to shortages after they occur. This creates a smoother overall customer experience and improves the shop's reliability.

Customers may not necessarily notice the technical system operating behind the scenes, and that is completely acceptable. What they will notice is the improvement in service consistency. They

are less likely to experience delays, cancellations, or interruptions caused by unavailable parts. Over time, this reliability strengthens customer trust and increases the likelihood that customers will continue returning to the shop for future services.

In highly competitive automotive service industries, customer satisfaction and reputation are extremely important. A repair shop known for completing services efficiently and on time is more likely to retain loyal customers and attract new ones through positive recommendations. Because of this, improving inventory management indirectly contributes to the long-term growth and reputation of the business within the local community.

- **Future**

Researchers:

The proposed study may also serve as a valuable reference for future researchers, particularly students and developers interested in inventory management systems, automotive business operations, and business process automation. Many small businesses, especially local automotive shops, continue to rely on manual or spreadsheet-based systems despite the operational problems these methods create. Because of this, there is still a strong need for studies that examine practical and affordable technology solutions specifically designed for small and medium-sized businesses.

This project provides a real-world example of how inventory management problems can be analyzed and addressed within an actual automotive repair shop environment. Unlike theoretical projects that focus only on hypothetical scenarios, this study is grounded in the actual operational experiences of D Speedway Auto Repair Shop. The challenges identified, including delayed

updates, disconnected records, inconsistent inventory counts, and inefficient stock monitoring, are real problems currently affecting the business.

Future researchers may use this study as a basis for developing similar systems for other industries such as retail stores, hardware shops, motorcycle repair businesses, or service-oriented enterprises that also struggle with inventory management. The study may also help researchers better understand the importance of centralized data management, automation, forecasting, and real-time inventory monitoring within small business operations.

In addition, the project contributes useful information regarding system usability and practical implementation. Many research studies focus heavily on technical development while giving less attention to whether the system is actually suitable for everyday business use. This study highlights the importance of designing systems that match real operational workflows and remain understandable for users with limited technical knowledge. Future developers may use these insights to create systems that are not only technically functional but also practical, user-friendly, and sustainable in real business environments.

The study may also serve as a reference for students conducting research related to forecasting systems, reorder planning, sales tracking, inventory automation, and decision-support systems. Since the project integrates several business management functions into a single platform, future researchers can explore ways to improve or expand these features further depending on the needs of other businesses or industries.

Overall, this study contributes both academically and practically by demonstrating how technology-driven inventory management systems can improve operational efficiency, reduce human error, strengthen decision-making, and support business growth within small automotive repair shop environments.

1.6 Definition of terms

This section provides definitions for the key terms and concepts used throughout this study. These definitions are intended to give the reader a clear and consistent understanding of the specific terminology referenced in the context of developing an inventory and sales management system for an automotive repair shop. Each term is explained in a straightforward manner to make the document accessible to a broad audience, including those who may not have a technical background.

Automotive Repair Shop

An automotive repair shop is a business establishment that provides maintenance and repair services for motor vehicles. The services offered typically include routine tasks such as oil changes, tire rotations, and battery replacements, as well as more complex procedures like engine diagnostics, brake system repairs, and transmission work. In the context of this study, the automotive repair shop serves as the primary setting for the proposed system. The shop relies on a steady supply of spare parts and consumables to carry out daily operations, and the efficient management of these resources is central to the problem being addressed.

Inventory Tracking

Inventory tracking refers to the process of monitoring and recording the quantity, location, and movement of parts and supplies within the shop at any given time. This includes updating stock counts whenever items are received from suppliers, used during repairs, or identified as damaged or expired. Accurate inventory tracking helps the shop avoid situations where needed parts are unavailable, prevents over-purchasing, and supports better financial planning. In this system, inventory tracking is one of the core functionalities, allowing staff to see what is currently in stock and managers to make well-informed purchasing decisions based on real data.

Parts Forecasting

Parts forecasting is the practice of using past transaction data and usage patterns to predict which parts will be needed in the coming days, weeks, or months, and in what quantities. Rather than simply reacting to stockouts when they happen, parts forecasting allows the shop to plan ahead and ensure that high-demand items are always available before they run out. This approach reduces delays in service delivery and helps the shop avoid both understock and overstock situations. The forecasting function in this system draws from historical sales records to generate estimates that guide future purchasing decisions.

Real-time Monitoring

Real-time monitoring refers to the capability of a system to display and update information immediately as events occur, without any delay or need for manual data entry after the fact. In the context of this study, real-time monitoring means that whenever a transaction is recorded — whether a part is sold, used in a repair, or received from a supplier — the system automatically reflects that change in the inventory count and other relevant records right away. This allows

shop owners and managers to always have an accurate, up-to-date picture of the shop's operations, which is especially useful when making quick decisions about restocking or managing ongoing repairs.

Reorder Tracking

Reorder tracking is a feature of the system that helps the shop keep tabs on the status of parts that have been flagged for reordering. When a part's stock level falls below its set minimum threshold, the system records this and allows the manager to initiate a reorder. The manager manually enters the quantity needed and selects the appropriate supplier, after which the system tracks the reorder through each stage — from pending to received. This feature replaces the need to rely on memory or informal notes, giving the shop a clear and organized record of what has been ordered, from whom, and when it is expected to arrive.

Prototype

In software development, a prototype is a functional early version of a system that is built to demonstrate its key features and test its core capabilities, without being fully deployed for actual use in a live environment. The prototype stage allows developers and stakeholders to evaluate whether the system works as intended, identify any problems or areas for improvement, and gather feedback before committing to full-scale development. In this study, the developed system is presented as a prototype — meaning it is operational and tested, but has not yet been formally deployed in a real automotive repair shop setting.

Stockout

A stockout occurs when a particular part or supply that is required to complete a repair is completely unavailable at the shop. This situation typically forces mechanics to pause or

postpone the job until the needed item can be sourced from a supplier, which can lead to delays in service, dissatisfied customers, and lost income for the business. Stockouts are often the result of poor inventory monitoring or a failure to reorder items in time. One of the primary goals of this system is to reduce the frequency of stockouts by keeping stock levels visible and sending timely alerts when supplies are running low

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Minimum Stock Threshold

The minimum stock threshold is a pre-set value that indicates the lowest acceptable quantity of a particular part before the system generates a restocking alert. Each part in the inventory can be assigned its own threshold value based on factors like how quickly it tends to sell, how long it takes to reorder, and how critical it is to daily operations. When the available stock of a part drops to or below this threshold, the system flags it so that the manager can take action. Setting appropriate thresholds is an important part of maintaining a well-functioning shop, as it provides an early warning before a stockout actually occurs.

Centralized System

A centralized system is a digital platform where all data related to the shop's operations is stored, managed, and accessed from a single location. Instead of maintaining separate spreadsheets, paper logs, or files for different aspects of the business — such as inventory counts, sales records, and supplier contacts — everything is kept together in one unified system. This makes it easier for authorized users to find information quickly, reduces the risk of data inconsistencies, and ensures that everyone working with the system is looking at the same up-to-date data. For a

small business like an automotive repair shop, having a centralized system can significantly reduce the time spent on administrative tasks.

Access Level

Access level refers to the set of permissions assigned to a particular user within the system, which determines what information they can view and what actions they are allowed to perform. Different roles within the shop may require different levels of access. For example, a regular staff member might be permitted to record sales and search for parts, but would not have the authority to modify pricing, delete records, or access financial summaries. The shop owner or manager, on the other hand, would typically have full access to all features and data. Defining clear access levels is important for protecting sensitive business information and ensuring that each user only interacts with the parts of the system relevant to their role.

Transaction

A transaction, in the context of this system, is any recorded event that involves the movement of parts or the exchange of payment. This includes sales made to customers, parts used during a repair job, and items received from suppliers. Each transaction is logged in the system with details such as the date, the item involved, the quantity, and the amount paid or received. These records serve as a foundation for inventory updates, sales reports, and financial tracking. Maintaining accurate transaction records is essential for understanding how the shop is performing and for making decisions based on reliable data rather than estimates.

Sales Report

A sales report is a summary document generated by the system that presents an overview of all transactions that occurred within a specified time period, such as a single day, a calendar month,

or an entire year. The report typically includes details on the total revenue earned, the individual parts or services that were sold, the quantities moved, and any breakdowns by payment method or category. Sales reports are a valuable tool for the shop owner and manager because they provide a clear picture of business performance over time, help identify which products are selling well, and highlight periods of high or low activity that may inform decisions about staffing, purchasing, or promotions.

Dashboard Metrics

The following terms describe the specific metrics and visual indicators available on the system's main dashboard. These features are designed to give the shop owner and manager a quick but meaningful overview of the business's current status and recent performance.

Daily Revenue — Parts vs. Labor

This metric displays a breakdown of the shop's total income for each day, separated into two categories: revenue earned from parts sales and revenue earned from labor or service fees. By viewing both figures side by side, the manager can quickly identify which days generate the most income and whether that income is driven primarily by the sale of parts or by the services rendered. This kind of visibility helps with scheduling, pricing, and understanding what types of work are most profitable for the shop on a day-to-day basis.

Monthly Revenue Trend

The monthly revenue trend presents a running view of the shop's total earnings across each month of the year. This metric allows the owner or manager to observe whether the business is growing over time, experiencing seasonal dips, or holding steady. By being able to compare performance month to month, management can identify patterns — for instance, noticing that

certain months are consistently slower — and use that information to plan staffing levels, promotions, or inventory orders more effectively. It provides a broader perspective on the business's financial health beyond just what happened in the current week.

Top 10 Products Sold

This metric ranks the ten best-selling parts or products within a selected time period, showing which items are in the highest demand among customers. Knowing which products move the fastest gives the shop practical guidance on where to focus restocking efforts and how to allocate storage space. It also helps with supplier negotiations, as the shop can anticipate volume purchases for consistently popular items. This feature turns raw sales data into actionable insight, making it easier to maintain the right stock at the right time.

Revenue Split — Parts vs. Labor

Similar to the daily revenue breakdown but viewed in aggregate, the revenue split metric shows the overall proportion of income that comes from parts sales compared to income from labor and services. This is typically displayed as a chart or percentage figure, giving the owner a quick, at-a-glance understanding of where the shop's money is coming from. If the split shifts unexpectedly — for example, if parts revenue drops significantly while labor stays the same — it may indicate a change in customer behavior or a problem with stock availability that warrants further investigation.

Stock Levels — Low Items Alert

This feature on the dashboard highlights parts whose current stock quantities have fallen at or below their designated minimum stock threshold. Rather than requiring the manager to manually review the full inventory list to check for low-stock items, the system surfaces this information

automatically on the main screen. The alert makes it easy to see at a glance which items need to be reordered before they run out entirely. This is directly connected to the concept of real-time monitoring, as the alerts update instantly based on the most recent transactions recorded in the system.

Payment Method Breakdown

The payment method breakdown metric shows how customers are paying for parts and services each month, specifically distinguishing between cash payments and digital payments made through GCash. This information is useful not only for accounting purposes but also for understanding customer preferences and ensuring that the shop is equipped to handle both payment types efficiently. If one method is being used significantly more than the other, the shop may want to adjust its processes accordingly — for example, ensuring that QR codes for GCash are always visible and accessible, or keeping adequate cash float on hand.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Sánchez-Luna, Rodríguez-Campos, and Rico-Chagollán (2025), in the study titled “Technology Transfer in Inventory and Order Management in Automotive Parts Stores” published in the Journal Applied Computing / Revista de Cómputo Aplicado, examined the implementation of technology-based inventory and order management systems within automotive parts stores. The study focused on how technological solutions can improve inventory control, order processing, and operational efficiency in automotive businesses that handle large quantities of parts and supplies on a daily basis.

The researchers explained that inventory and order management are critical components of automotive operations because businesses depend heavily on the availability of parts to maintain continuous service delivery. In automotive environments where transactions occur frequently and inventory movement happens rapidly, businesses require systems capable of managing stock information accurately and efficiently. The study highlighted that even businesses already using digital tools may still experience operational inefficiencies when inventory systems are not fully integrated into daily workflows or when inventory data is not updated consistently in real time.

According to the study, common inventory-related problems in automotive parts stores include discrepancies between recorded and actual stock levels, delays in updating inventory information, inefficient order monitoring, and difficulties in processing transactions accurately. These problems can negatively affect operational performance by causing stock shortages, delayed

customer service, inaccurate reporting, and unnecessary financial losses. The researchers emphasized that inventory management systems must not only store data digitally but must also support real-time monitoring and efficient workflow integration in order to be fully effective.

The study further discussed the importance of improving inventory visibility and streamlining order management processes. When inventory systems are capable of updating stock information immediately after transactions occur, businesses are better able to monitor inventory conditions, identify low-stock items earlier, and make more informed purchasing decisions. Real-time inventory visibility also allows businesses to respond more quickly to customer demands and minimize delays caused by unavailable parts.

To address these operational challenges, the researchers proposed the implementation of an improved web-based inventory and order management system designed to enhance inventory accuracy, improve monitoring processes, and support more efficient handling of orders and stock movement. The study emphasized that technology-based systems can significantly reduce the administrative burden associated with manual inventory management while improving the reliability and accessibility of business records.

Another important point raised in the study was that technology adoption alone does not automatically guarantee operational improvement. The researchers explained that inventory systems must be properly aligned with the actual workflow and operational needs of the business. Systems that are poorly integrated into daily operations may still produce inaccurate records and inefficient processes even when digital tools are being used. Because of this, the study highlighted the importance of designing systems that support how automotive businesses naturally operate rather than forcing users to adjust to overly complex software structures.

The findings of Sánchez-Luna et al. (2025) are highly relevant to the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop because the operational problems discussed in the study closely resemble the issues currently experienced by the shop. Similar to the automotive parts stores examined in the research, D Speedway currently relies on manually maintained spreadsheets to handle inventory and sales tracking. This setup creates challenges related to stock accuracy, inventory visibility, delayed updates, and inefficient monitoring processes.

The study strongly supports the need for real-time inventory tracking within the proposed system. At present, D Speedway has no centralized platform capable of automatically updating stock records whenever parts are sold or used during repairs. As a result, staff members sometimes encounter discrepancies between recorded inventory and actual stock availability, which can delay repairs and disrupt customer service operations. By implementing real-time inventory updates and centralized record management, the proposed system aims to eliminate many of these inefficiencies.

Additionally, the study reinforces the importance of integrating inventory systems directly into operational workflows. The proposed Smart Inventory and Parts Planning System is specifically designed around the actual working environment of D Speedway. Instead of forcing staff members to adapt to complicated processes, the system is intended to simplify inventory management by automating stock updates, monitoring low-stock items, generating reports, and supporting reorder planning in a way that fits naturally into the shop's daily operations.

The study also supports the importance of improving inventory visibility and order management efficiency through automation. Features planned for the proposed system — such as automated

inventory monitoring, stock alerts, forecasting, and reorder recommendations — directly align with the technological improvements discussed in the research. These functionalities are intended to reduce manual administrative work, improve inventory accuracy, and help the shop respond more effectively to customer needs and changing demand patterns.

Overall, the study conducted by Sánchez-Luna et al. (2025) provides strong support for the development of the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop. The research demonstrates that effective inventory and order management systems must provide real-time monitoring, accurate stock control, and workflow integration in order to improve operational performance within automotive businesses. The findings reinforce the value of implementing a centralized and technology-driven system capable of improving inventory accuracy, operational efficiency, and overall service reliability in automotive repair environments.

According to Agboola, Malgwi, Mahmud, and Oguntoye (2022), inventory management is one of the most important operational areas in small and medium enterprises because it directly affects efficiency, cost control, and customer satisfaction. In their study titled *Development of a Web-Based Platform for Automating an Inventory Management of a Small and Medium Enterprise*, published in the *FUDMA Journal of Sciences* (Vol. 6, No. 5, ISSN: 2384-5821), the researchers focused on creating a computerized inventory management platform designed to replace the traditional manual methods that were still being used by Sangkemi Global Nigeria Limited, a small and medium enterprise in Adamawa State, Nigeria. The study started from a

simple but important premise: the way most small businesses track their inventory is not working, and the longer they wait to change it, the worse the consequences become.

The study explained that a large number of small businesses continue to rely on spreadsheets, handwritten records, and disconnected files to monitor inventory and sales transactions. On the surface, these methods seem like they should be enough — after all, a spreadsheet can hold a lot of data and most people already know how to use one. But Agboola et al. (2022) pointed out that the problem is not the tool itself, it is what happens when you ask that tool to do more than it was designed for. As a business grows and transactions increase, the spreadsheet approach becomes harder and harder to sustain. What started as a manageable file becomes a collection of files, each one maintained by different people at different times, each one drifting slightly away from the others until no single record can be fully trusted. The researchers were clear that these are not problems unique to one business or one industry — they are the predictable outcome of using manual systems in environments that have outgrown them.

Agboola et al. (2022) also pointed out the financial dimension of poor inventory management that often gets overlooked in discussions about small businesses. The researchers cited that inventory costs for businesses still using manual processes can range anywhere from four percent to ninety percent of total business expenses, depending on the industry and the extent of the errors involved. That is an enormous range, and even at the lower end it represents a significant portion of operational spending that is being influenced — often negatively — by inaccurate and outdated records. For a small business like D Speedway where parts inventory is both the primary cost and the primary driver of service delivery, sitting anywhere on that range without knowing it is a real financial risk. The study framed this not as a scare tactic but as a factual

context for why manual inventory management is not a neutral choice — it is a choice with measurable financial consequences that compound over time.

The researchers further discussed how inaccurate inventory records affect overall business operations beyond just the numbers. When stock information cannot be trusted, businesses lose the ability to make informed purchasing decisions. The owner ends up guessing — ordering more of something because they are not sure how much is left, or waiting too long to reorder because the records suggest more stock than actually exists. Both of those outcomes carry costs. Agboola et al. (2022) noted that overstocking increases storage costs and ties up capital that could be used elsewhere in the business, while stock shortages delay operations, disrupt the workflow, and push customers toward frustration and eventually toward competitors. The shop does not have to be doing something wrong for these problems to develop — they develop naturally when the system for tracking inventory is not keeping up with the pace of the business. That is exactly what is happening at D Speedway, where the research team observed that the staff and owner are not careless people. They are capable people working in a system that was not designed to handle what they are putting it through.

One of the more specific findings that Agboola et al. (2022) documented was what happens to staff time when inventory records are unreliable. Because the records cannot be trusted, employees stop using them as the primary reference and start doing things manually that the records were supposed to handle. Instead of checking a system, someone walks to the storage area. Instead of pulling a report, the owner sits down with a calculator. Instead of trusting the numbers in a file, a staff member cross-references two or three files to try to figure out what the actual situation is. None of that is productive time. It is time spent compensating for a broken

system, and it happens every single day across every member of staff who touches inventory data. The researchers made the point that this kind of invisible time loss is one of the most significant costs of manual inventory management — significant because it is real, and invisible because it never shows up as a line item on any report.

To address these challenges, Agboola et al. (2022) developed a web-based inventory management platform capable of automating inventory monitoring and record keeping. The system was built at Sangkemi Global Nigeria Limited using the Iterative Waterfall Model, a development methodology that moves through structured phases but allows for corrections at each stage before moving forward, which reduces the risk of building something that does not actually solve the problem. The researchers chose a web-based architecture specifically because it could be accessed from any device with an internet connection, did not require special software to be installed on individual machines, and could support multiple users at the same time without creating the kind of data conflicts that happen when several people are working in the same spreadsheet file. The system stored all inventory data in a centralized digital database, updated stock records automatically whenever a transaction was recorded, and gave users fast access to current information without requiring them to dig through files or rely on memory.

The system was built around three core functional modules. The first was a sales management module, which allowed staff to record transactions through the system directly so that every sale was immediately logged and the corresponding inventory count was automatically adjusted. This removed the manual step of updating a stock record separately after each sale — a step that Agboola et al. (2022) identified as one of the primary points where errors enter the system. The second was a product tracking module that displayed real-time stock levels for every item in the

inventory, so staff could check availability instantly from wherever they were working rather than walking to the storage area or opening a file that might be out of date. The third was a user management module that controlled what each type of user could access and do within the system, with administrators having full access and sales staff seeing only the functions relevant to their role. This structure protected sensitive business data while keeping the interface clean and manageable for employees who only needed to do basic tasks.

Another important contribution of the study was its emphasis on real-time visibility as a practical operational tool, not just a technical feature. Agboola et al. (2022) explained that real-time monitoring changes the way a business responds to its own inventory situation. When stock levels update instantly with every transaction, the owner does not need to schedule a stock check or wait for a report to know what is happening — the information is there whenever it is needed. This means that low-stock situations can be caught while there is still time to reorder through normal channels at standard prices, rather than being discovered only when a mechanic is already mid-repair and the part they need has already run out. For D Speedway, where the research team documented cases of exactly that kind of mid-repair discovery, this distinction is not minor. It is the difference between a planned, organized purchasing decision and a panicked rush order paid at a premium.

The study also highlighted the importance of centralized data management in a way that goes beyond just keeping things organized. Agboola et al. (2022) argued that when data is scattered across separate files maintained by different people, the real problem is not just that the files are inconsistent — it is that there is no longer a reliable single source of truth for the business. When two records contain different numbers for the same item and there is no way to determine which

one is correct, every decision based on either of those records carries an unacknowledged margin of error. The owner of the business may not even know the margin is there. They are reading a number and treating it as fact when it is actually a guess in disguise. Centralizing all information into one connected system eliminates that ambiguity. There is one number. It updates automatically. Everyone in the business is looking at the same data at the same time. Decisions made from that data are decisions made from a foundation that can actually be trusted.

The findings of the study confirmed what the researchers had set out to demonstrate: implementing the automated web-based platform significantly improved efficiency, reduced workload on staff, and increased the accuracy of inventory records at Sangkemi Global Nigeria Limited. After the system was deployed, the owner could monitor all business processes through a single platform without relying on physical book entries or manual calculations. The record-keeping burden that had previously fallen on individual employees to maintain correctly and on time was handled by the system automatically. Reports that had required manual compilation were generated instantly from stored data. The study concluded that digital inventory management systems are no longer optional for growing businesses — they have become necessary tools for maintaining operational stability in environments where transaction volumes are too high and too complex for manual methods to handle reliably.

The study of Agboola et al. (2022) is strongly relevant to the proposed Smart Inventory and Parts Planning System for D Speedway because the two situations are genuinely comparable in ways that go beyond surface-level similarity. Both involve small businesses that sell or use physical parts as the core of their operations. Both are growing faster than their manual systems can keep up with. Both are experiencing the same specific set of consequences — inaccurate records,

unreliable stock data, time-consuming manual reporting, and purchasing decisions made without a clear view of what is actually on hand. The researchers at Agboola et al. (2022) documented those problems in a Nigerian SME. The research team for this project documented them at D Speedway through site visits and staff interviews. The problems are the same because the cause is the same: manual systems being asked to do more than they were designed to handle.

The solution that Agboola et al. (2022) developed — a centralized web-based platform that automates stock updates, provides real-time visibility, consolidates data into one system, and generates reports without manual effort — is also the same type of solution being built for D Speedway. The proposed Smart Inventory and Parts Planning System addresses exactly the problems identified in this study: disconnected records are replaced by a centralized database, manual stock updates are replaced by automatic deductions after every recorded transaction, delayed report generation is replaced by real-time sales summaries, and reactive stockout discoveries are replaced by proactive low-stock alerts that give the shop time to reorder before a shortage disrupts a repair. The design philosophy behind both systems is identical — reduce the manual work, automate the record keeping, and give the owner the visibility they need to run the business well.

Additionally, the emphasis on real-time inventory visibility in the Agboola study strongly supports the forecasting and reorder recommendation features planned for the proposed system. Real-time visibility is not just about knowing what is currently on the shelf — it is the foundation that makes forward-looking planning possible. If the stock data is accurate and current, the system can analyze how quickly certain parts are being used and project when they will need to be restocked. If the purchase history is recorded correctly, the system can suggest

how much to order and which supplier to contact. None of those features work properly if the underlying data is unreliable. Since D Speedway handles multiple vehicle repairs daily, having accurate and updated inventory data is essential for ensuring that mechanics always have the necessary parts available when needed. Through automation, the proposed system will reduce manual workload, minimize stock-related disruptions, improve operational efficiency, and help the business continue growing without the administrative problems caused by spreadsheet-based inventory management.

The study also supports the importance of adopting technology early, while the business is still at a manageable scale. D Speedway has only been operating for one year, but the inventory and tracking issues are already becoming difficult to manage. Agboola et al. (2022) made the same observation about the businesses they studied — that relying on manual inventory systems becomes increasingly unsustainable as transaction volume and inventory complexity continue to grow, and that the longer a business waits to address the problem, the more embedded the broken habits become and the harder the transition gets. Implementing an automated inventory and parts planning system at this stage of D Speedway's growth is not premature. It is the right moment — while the scale is still manageable, while the data that needs to be migrated is still relatively clean, and while the staff can be trained on a new system without the disruption that comes from making changes in the middle of a fully scaled operation. The findings of Agboola et al. (2022) confirm that this kind of early action is not just practically smart but strategically necessary for a business that intends to keep growing.

Ivan and Oetama (2024), in their study titled “Inventory Management System Using Economic Order Quantity and Reorder Point” published in *G-Tech: Jurnal Teknologi Terapan*, examined the inventory management practices of an automotive workshop that relied entirely on Microsoft Excel for tracking parts and stock levels. The researchers found that while spreadsheets were initially manageable for recording inventory data, they became increasingly inefficient and unreliable as the workshop expanded its operations and handled a wider range of automotive parts. Since all inventory calculations depended on manual updates, the workshop struggled with maintaining accurate stock records and monitoring inventory movement consistently.

The study explained that the workshop had no automated mechanism for identifying when inventory levels were becoming critically low. Instead, stock monitoring depended heavily on staff manually checking records and noticing shortages before they disrupted operations. Safety stock limits were only applied to a small number of fast-moving items, while the majority of inventory lacked any organized monitoring process. Because of this, stockouts frequently occurred without warning, leading to delays in repairs, interrupted workflow, and difficulties in maintaining smooth daily operations. The researchers emphasized that inaccurate inventory planning and poor stock monitoring can negatively affect the entire supply chain of a workshop, especially when needed parts are unavailable at the exact moment repairs are being performed.

The researchers further discussed how manual inventory systems create long-term operational inefficiencies. Since the workshop depended on Excel files that required constant manual updates, inventory accuracy relied entirely on the consistency and attentiveness of staff members. In busy automotive environments where employees are focused primarily on servicing vehicles and assisting customers, updating inventory files accurately and immediately often becomes a

secondary priority. As a result, discrepancies between recorded stock levels and actual physical inventory gradually increase over time. The study noted that once staff lose confidence in the accuracy of inventory records, they begin relying more on manual stock checking rather than the system itself, which defeats the purpose of maintaining inventory records in the first place.

To address these challenges, Ivan and Oetama (2024) developed a computerized inventory management application that integrated two established inventory control methods: Economic Order Quantity (EOQ) and Reorder Point (ROP). The Economic Order Quantity method was designed to determine the most cost-efficient quantity of parts to order during each restocking cycle. EOQ balances the cost of ordering inventory too frequently against the cost of storing excessive stock, helping businesses minimize unnecessary expenses while maintaining sufficient inventory availability.

The Reorder Point method, on the other hand, focused on identifying the exact stock level at which a new order should automatically be triggered. The system calculated reorder points based on the average usage rate of a part and the lead time required for suppliers to deliver new stock. By combining EOQ and ROP, the system transformed inventory management from a reactive process into a proactive one. Instead of waiting until parts completely ran out before placing orders, the workshop could now identify low-stock situations early enough to reorder inventory before operations were affected.

Another important aspect of the study was the use of Rapid Application Development (RAD) during system development. The researchers explained that RAD allowed the system to be developed in smaller stages with continuous feedback from actual workshop users. Rather than designing the system entirely based on assumptions, the developers regularly consulted

workshop staff throughout the process to ensure that the final product matched the real operational needs of the business. This approach improved usability because the people who would eventually use the system were directly involved in shaping how it functioned.

The results of the study showed that the developed system was highly effective and well accepted by its users. User Acceptance Testing conducted with actual workshop personnel produced an average satisfaction rating of 98.44 percent among administrative users and 97 percent among general users. These findings are significant because they demonstrate that the system was not only technically functional but also practical and easy to use in a real automotive workshop environment. The researchers emphasized that user-friendly design is essential in automotive businesses where employees are not software specialists and require systems that support their work without adding unnecessary complexity.

The study of Ivan and Oetama (2024) is highly relevant to the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop because both studies address nearly identical operational challenges. Similar to the workshop discussed in the study, D Speedway currently relies on manually maintained Excel spreadsheets for managing inventory and recording sales transactions. This setup creates many of the same problems identified by the researchers, including inaccurate stock records, disconnected files, lack of real-time inventory visibility, delayed reporting, and the absence of automatic stock alerts.

The issue of reactive inventory management described in the study closely reflects the current situation at D Speedway. At present, the shop often discovers inventory shortages only after a needed part is already unavailable, forcing mechanics to pause repairs while replacement parts are sourced from suppliers. This not only delays service completion but also creates additional

operational costs and negatively affects customer satisfaction. The implementation of automated reorder alerts and inventory forecasting within the proposed system directly addresses these problems by helping the shop identify low-stock items before shortages disrupt daily operations.

The EOQ and ROP concepts presented in the study also provide valuable guidance for the forecasting and reorder recommendation features planned for D Speedway's proposed system. By analyzing historical sales data and monitoring inventory movement, the system can recommend appropriate reorder quantities and determine when new stock should be ordered. This allows the shop to maintain adequate inventory levels without overstocking parts that may not be needed immediately.

Additionally, the study strongly supports the importance of usability in system design. Like the workshop employees in Ivan and Oetama's research, the staff at D Speedway are primarily automotive service workers rather than technology specialists. Because of this, the proposed system must remain simple, intuitive, and easy to operate in a busy repair shop environment. The high User Acceptance Testing results achieved in the study demonstrate that inventory systems can be both technically advanced and user-friendly at the same time when designed around the real needs of workshop staff.

Overall, the study provides both a technical foundation and a practical real-world example that supports the development of the proposed Smart Inventory and Parts Planning System for D Speedway. It demonstrates that automated inventory management, proactive stock monitoring, reorder planning, and user-centered system design can significantly improve operational efficiency, inventory accuracy, and service reliability within automotive workshop environments.

Surbakti and Selly (2023) conducted a study published in the International Journal of Scientific Research and Management (Vol. 11, No. 05, e-ISSN: 2321-3418) about designing a web-based information system for a motorcycle spare parts store in Depok City, Indonesia called Bintang Multi Sparepart Store. The store has been operating since 2017 and runs with only three full-time employees who handle everything from serving customers to managing stock to closing the books at the end of each day. Its main customers are owners of nearby repair workshops and regular buyers who come in looking for specific motorcycle parts. That customer profile is worth noting because it is not very different from D Speedway, where mechanics are the ones pulling parts from inventory daily and the business depends on having those parts available and accounted for at all times. The study is one of the more relevant pieces of literature for this project because it deals with an actual automotive parts business — not a hypothetical scenario — and the problems it found at Bintang Multi are the same ones this research team found at D Speedway when they visited the shop and talked to the staff.

Before getting into what the study built, it helps to understand what the store was dealing with before any system existed. The way Bintang Multi handled its daily operations was entirely manual. When a customer came in to buy a part, the employee would first have to physically walk to the storage area to check if the item was available. The storage area was described in the study as being considerably far from the customer service area, which meant that every single stock check wasted time — both the employee's and the customer's. If the part was available, the employee would go back, write out a receipt for the customer, and then record the transaction a second time in a physical sales book, noting down the quantity, item type, item code, and the

price. That double entry process happened for every sale throughout the day. At closing time, the owner would sit down with all those sales book records and manually calculate the day's total using a calculator. The same thing happened at the end of each month with a monthly summary. On the surface it sounds manageable, but Surbakti and Selly (2023) pointed out that errors happened constantly under this setup — not because the employees were sloppy, but because the conditions of manual record-keeping almost guarantee mistakes over time. The book got stained or torn, handwriting became hard to read, numbers were transposed, entries were missed entirely. These are the kinds of small errors that add up silently until the records and the actual cash no longer match, and the owner has to recount everything from scratch to find where it went wrong.

The stock fulfillment process had its own set of problems separate from the sales recording. There was no tracking system that showed current stock levels. No alert went out when a product was running low. The only way anyone knew a part had run out was when an employee went to the storage area for an order and found the shelf empty, at which point they would report it to the owner. The owner would then manually write up a list of items to reorder and contact the supplier. Because this entire process depended on someone noticing and reporting a shortage at the right time, parts sometimes stayed out of stock for extended periods simply because no one caught it before it became a problem. The researchers specifically noted that stock was left unfulfilled for long stretches due to the owner forgetting to reorder — not out of negligence, but because there was nothing in the system reminding them that it needed to be done. This is exactly the kind of situation D Speedway deals with. The shop has had mechanics start a repair only to discover partway through that the part they needed was already gone. That kind of disruption happens because there is no mechanism in place to catch a low-stock situation before it becomes a zero-stock situation.

The financial reporting problem at Bintang Multi was perhaps the most telling detail in the study. Because the profit and loss calculation depended on reconciling the sales book with the actual cash on hand, and because that reconciliation was frequently off due to recording errors, the owner had stopped trying to calculate actual profit from the records. Instead, the owner applied a rough estimate of ten percent of total sales as the assumed profit for the period. That is not a business decision — that is a workaround for a broken system. The owner of a business with daily transactions and a storage full of parts had no reliable way to know what the shop was actually earning because the records could not be trusted to give the right answer. This connects directly to D Speedway, where the owner has expressed the same frustration: getting a clear read on how the business is doing requires hours of manual work sorting through spreadsheet files, and even after all that effort, the figures are not always reliable because the files are updated by multiple people and do not always stay consistent with each other.

Surbakti and Selly (2023) developed their solution using the System Development Life Cycle methodology, which guided the project through four structured phases: planning, analysis, design, and implementation. What makes their approach worth highlighting is that they did not start with the system — they started with the business. Before any design work began, the researchers visited the store in person, observed the operations directly, and interviewed the owner to understand not just what the problems were but how the day-to-day work actually flowed. That observation and interview process is what allowed them to design a system that fit the store's actual operations rather than something the owner and staff would have to adapt themselves to. This is the same approach taken in this project. The research team visited D Speedway, watched how the shop operated, and talked to the owner and the staff before any feature decisions were made. The literature consistently supports this kind of needs-first design approach, and Surbakti

and Selly's study is one of the clearest examples of why it matters. The system they built worked because it was shaped by the people who would use it, not just by technical assumptions about what an inventory system should include.

The system itself was built using PHP as the programming language and MySQL for the database, delivered through a web-based interface. The decision to go web-based was a practical one: web applications do not require special software to be installed on individual computers, can be accessed from any device with an internet connection, and allow multiple users to work in the system at the same time without conflicts. All the product data — codes, names, categories, sizes, brands, stock quantities, prices, and descriptions — was stored in a centralized database that connected directly to the transaction process. This meant that every time a sale was recorded, the system automatically updated the stock count for whatever was sold without anyone having to do it separately. That automatic update is the core of what makes a system like this different from a spreadsheet. With a spreadsheet, the transaction and the inventory update are two separate manual steps, and any time someone skips or delays the second step, the records start drifting away from reality. With the system Surbakti and Selly (2023) built, the two things happen together — one action in the system handles both. Daily and monthly sales reports were also generated automatically from the stored data, which meant the owner no longer had to sit down at the end of each day and manually compile and reconcile figures from a record book. The information was already organized and available whenever it was needed.

The researchers also removed several manual steps from the stock fulfillment process. Because the system showed current stock levels at all times, employees no longer needed to walk to the storage area every time a customer placed an order. They could check availability directly

through the system from wherever they were standing. The reordering process also became more organized because the owner could see at a glance which items were running low without waiting for an employee to report a shortage. This shift from a reactive approach — finding out something is out of stock after the fact — to a more visible, proactive one is exactly what D Speedway needs. Right now the shop only knows a part is gone when someone goes looking for it and comes back empty-handed. A system that shows stock levels in real time removes that blind spot entirely.

The testing phase covered the system from multiple angles. Unit testing, integration testing, and system testing were all conducted using a black box approach where users were given scenario cases to run through the system so that any errors or weaknesses could be caught before the system was finalized. Acceptance testing was carried out using usability testing that measured five specific dimensions: learnability assessed how quickly users could figure out how to use the system for the first time; efficiency measured how fast they could complete tasks once they were familiar with it; memorability tested whether users could come back to the system after a break and still use it without relearning everything; error rate tracked how many mistakes users made during operation; and satisfaction was measured through the System Usability Scale questionnaire, which is a widely used standardized tool for evaluating whether a system is genuinely easy to use in real conditions. The system passed all of these tests, and the study concluded that the designed information system fully met the user's needs in terms of both functionality and usability.

After the system was implemented, the changes at Bintang Multi Sparepart Store were concrete. The owner could monitor all business processes — product availability, transaction records, and

sales summaries — through one platform instead of piecing together information from a book, a calculator, and a storage room. Real-time sales data was available at any point during the day without requiring the owner to stop and do manual calculations. The double entry problem was gone because the system handled the record and the stock update together. The reports that used to take hours to compile were generated by the system instantly. The study concluded that the web-based information system successfully replaced the manual approach and gave the business the data reliability and visibility it had been missing.

Looking at this study alongside the situation at D Speedway, the connection is hard to miss. Bintang Multi and D Speedway are different businesses in different countries, but the problems they were dealing with are described in almost the same terms: manual records that could not keep up with the pace of daily operations, stock levels that nobody could verify without checking physically, reports that required hours of manual effort to produce and still were not fully reliable, and an owner trying to make business decisions without a clear and trustworthy picture of what was actually happening. Surbakti and Selly (2023) showed that a web-based system built around the real workflow of a spare parts business resolves all of these things at once. The features that fixed Bintang Multi — centralized product data, automatic stock updates tied to sales, real-time availability checking, and automated report generation — are the same features being built into the Smart Inventory and Parts Planning System for D Speedway. This study does not just support the argument that a system like this is a good idea. It is evidence that it works, in a business very much like the one this project is designed for.

2.2 Local Studies

Barcelona, Gungon, Laylo, Panganiban, Sultan Amanollah, and Altarejos (2025) conducted a study that looked into how auto supply businesses in Candelaria, Quezon, Philippines manage their inventory. Their study involved nine business owners and managers as respondents and made use of a descriptive-correlational research design to better understand how inventory practices affect the overall performance of auto supply businesses in the area.

The study found that many auto supply businesses in the locality are still relying on outdated ways of managing their inventory, particularly the use of spreadsheets. These traditional methods are not only time-consuming but are also prone to human errors, often leading to inaccurate data, overlooked stocks, and internal losses that negatively affect daily business operations. This kind of situation makes it clear that there is a strong need for a more reliable and organized system, such as an automotive inventory system, that can handle stock management more accurately and efficiently. The researchers also identified two major problems that prevent these businesses from managing their inventory effectively.

The first is the cost of compliance, which includes securing permits, licenses, and certifications required by local government units. The second is the unpredictable movement of stock levels, which makes it difficult to maintain a stable and organized inventory. Together, these issues lead to unnecessary expenses, excessive stock buildup, and frequent shortages that directly affect the profitability and smooth operation of auto supply businesses. An automotive inventory system can help lessen these burdens by providing real-time tracking of stock levels, automated alerts for low inventory, and organized records that reduce the likelihood of human error. To help address these concerns, the study recommended the adoption of a Just-in-Time (JIT) inventory

approach, where businesses order stocks only when needed rather than keeping large amounts on hand.

This strategy helps lower storage costs, minimizes the risk of overstocking, and prevents automotive parts from becoming outdated or obsolete before they are sold or used. An automotive inventory system supports this kind of approach by giving business owners accurate and up-to-date information on their current stock, making it easier to decide when and what to reorder. Based on the findings of this study, it is evident that an automotive inventory system is something that local auto supply businesses genuinely need. The challenges that were pointed out — such as keeping track of stocks manually, dealing with incorrect data, managing unpredictable inventory levels, and not having access to real-time updates — are all things that an automotive inventory system can directly help with. Having a system that automatically monitors stocks, produces up-to-date inventory reports, and keeps the ordering process more organized can make a big difference in how auto supply businesses operate, helping them become more efficient, more accurate, and better equipped to handle the demands of running a business.

Bergonia (2021), in the study titled “The Inventory Management Practices of the Retail Automotive Industry in Pangasinan” published in the Ascendens Asia Journal of Multidisciplinary Research Conference Proceedings, examined the inventory management practices being implemented by automotive businesses in Pangasinan, Philippines. The study focused specifically on how businesses in the automotive retail industry manage procurement, storage, stock monitoring, inventory information systems, and inventory-related decision-making

processes. The research highlighted the growing importance of inventory management in automotive businesses, where daily operations heavily depend on the availability of vehicle parts, tools, supplies, and other materials necessary for repair and maintenance services.

The study explained that inventory management is one of the most important operational functions in automotive businesses because it directly affects workflow efficiency, customer satisfaction, and financial performance. Automotive businesses constantly deal with different types of parts that vary in demand, availability, pricing, and supplier lead time. Because of this, maintaining accurate inventory records and monitoring stock levels consistently are necessary to avoid disruptions in service operations. According to the study, businesses that fail to manage inventory effectively often experience operational inefficiencies such as stock shortages, overstocking, delayed repairs, inaccurate reporting, and unnecessary expenses.

Bergonia (2021) found that while many automotive businesses had established procurement and storage procedures, their inventory information systems were only considered “fairly effective.” This finding indicated that many businesses still relied heavily on traditional or partially manual methods of managing inventory data. Inventory records were often maintained through handwritten logs, spreadsheets, or disconnected files that required constant manual updating by employees. Although these methods allowed businesses to maintain basic inventory operations, they lacked the reliability, speed, and accuracy needed to support growing automotive businesses effectively.

The study emphasized that manual inventory systems create several long-term operational challenges. Since inventory updates depend entirely on employees manually recording every transaction, businesses become highly vulnerable to human error. Mistakes such as forgotten

updates, duplicate entries, incorrect stock counts, and delayed recording of inventory movement become increasingly common, especially in busy automotive environments where staff members prioritize repairs and customer service over administrative tasks. Over time, these small inaccuracies accumulate and cause the inventory records to no longer match the actual physical stock available in storage.

Another major issue identified in the study was the difficulty businesses faced in maintaining real-time inventory visibility. Because inventory information was not updated automatically, business owners and employees often had no immediate way of knowing the exact quantity of parts available at any given moment. In many cases, staff members had to physically inspect storage areas or manually cross-check several files just to confirm whether a particular part was still in stock. The study explained that this process consumed unnecessary time, reduced productivity, and created delays in customer service operations.

The lack of real-time visibility also negatively affected decision-making processes within automotive businesses. Without accurate and updated inventory data, business owners struggled to determine which parts needed restocking, which products were moving quickly, and which items were remaining unsold for long periods. As a result, businesses became more vulnerable to both understocking and overstocking problems. Understocking created service delays because mechanics could not complete repairs immediately when parts were unavailable, while overstocking tied up financial resources in inventory that was not moving efficiently.

Bergonia (2021) further explained that inventory-related problems have a direct impact on customer satisfaction and business reputation. In automotive businesses, customers expect repairs and maintenance services to be completed within reasonable timeframes. However, when

necessary parts are unavailable due to poor inventory monitoring, repairs are delayed and customers are forced to wait longer than expected. Repeated delays caused by stock shortages can negatively affect customer trust and reduce the likelihood of repeat business. The study emphasized that efficient inventory management is not only a technical or administrative concern but also a critical component of delivering reliable customer service.

The study also highlighted the importance of centralized inventory information systems. According to the researcher, businesses that maintain separate files for inventory, procurement, and sales records often experience inconsistencies between records because information is updated independently by different employees at different times. This fragmented structure makes reporting more difficult and increases the possibility of conflicting data. Bergonia noted that centralized systems help solve these issues by consolidating all inventory-related information into a single platform that can be accessed and updated consistently.

Another important point discussed in the study was the growing need for businesses in the automotive industry to adopt computerized inventory management systems as operations continue to expand. Bergonia explained that while manual systems may initially seem manageable for small businesses, they eventually become inefficient as inventory volume, transaction frequency, and customer demand increase. Businesses that continue relying on spreadsheets and manual records often struggle to scale operations efficiently because their administrative systems cannot keep up with the growing complexity of inventory management.

The study recommended that automotive businesses adopt more advanced inventory information systems capable of automating stock monitoring, improving record accuracy, generating faster reports, and supporting better decision-making. The researcher emphasized that digital inventory

systems can reduce administrative workload, improve operational efficiency, and provide business owners with more reliable information for planning and purchasing decisions. Through automation, businesses can minimize inventory-related disruptions and improve their ability to respond quickly to customer needs.

The findings of Bergonia (2021) are highly relevant to the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop because the operational challenges identified in the study closely mirror the current problems being experienced by the shop. Similar to the businesses discussed in the research, D Speedway currently relies on manually maintained Excel spreadsheets to record inventory and sales transactions. This setup creates many of the same difficulties identified by Bergonia, including inaccurate inventory records, disconnected files, delayed updates, inefficient reporting, and the absence of real-time stock monitoring.

The issue of fragmented and disconnected inventory records discussed in the study is especially relevant to D Speedway's current operations. At present, the shop manages sales records, inventory information, and supplier details through separate spreadsheets that are updated manually by staff members. Since these files are not automatically connected to one another, inventory counts can easily become inconsistent whenever updates are delayed or forgotten. This problem often forces staff to manually verify stock levels through physical inspection rather than relying on the records themselves, resulting in wasted time and reduced operational efficiency.

The study's discussion on real-time inventory visibility also strongly supports the objectives of the proposed system. D Speedway currently has no automated mechanism capable of instantly updating inventory records whenever a part is sold or used during repairs. Because of this, the

shop sometimes discovers inventory shortages only after mechanics have already started a repair job. These situations delay operations, disrupt workflow, and negatively affect customer experience. The proposed Smart Inventory and Parts Planning System directly addresses this issue by implementing automatic inventory updates and low-stock alerts that notify staff before shortages occur.

Additionally, the recommendation for adopting centralized computerized inventory systems aligns directly with the goals of this project. The proposed system for D Speedway aims to replace the shop's fragmented spreadsheet-based process with a unified platform capable of handling sales recording, inventory monitoring, stock alerts, forecasting, and report generation within a single system. By centralizing inventory and sales data, the proposed system will improve data consistency, reduce administrative workload, and provide the shop owner with more accurate information for decision-making.

The study also supports the importance of inventory forecasting and proactive stock planning. Bergonia emphasized that businesses need reliable inventory data to make effective purchasing decisions and avoid stock-related disruptions. This directly relates to the forecasting and reorder recommendation features planned for the proposed system, which aim to help D Speedway anticipate future inventory demand based on historical sales patterns. By identifying which parts are used most frequently and monitoring inventory movement in real time, the system can help the shop reorder parts before shortages occur and reduce the risk of delayed repairs.

Furthermore, the study reinforces the idea that inventory management systems must support the actual workflow of automotive businesses rather than forcing businesses to adapt to complicated software structures. Automotive repair shops operate in fast-paced environments where

mechanics and staff members are focused primarily on servicing vehicles and assisting customers. Because of this, inventory systems must remain practical, user-friendly, and easy to operate even for users with limited technical knowledge. This principle is reflected in the design objectives of the proposed Smart Inventory and Parts Planning System, which aims to simplify inventory management while minimizing additional workload for the staff.

Overall, the study conducted by Bergonia (2021) provides strong evidence supporting the development of the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop. The research demonstrates that ineffective inventory information systems can significantly affect inventory accuracy, operational efficiency, reporting reliability, and customer satisfaction within automotive businesses. At the same time, the study highlights how centralized and computerized inventory management systems can improve business operations by automating stock monitoring, reducing manual errors, improving inventory visibility, and supporting more effective decision-making. These findings reinforce the importance of implementing a reliable inventory and parts planning system specifically designed to address the operational needs of D Speedway as it continues to grow.

Roque et al. (2023) conducted a study titled “Development of an Inventory and Service Management System for an Auto Repair Shop Using Agile Methodology,” which was published in Philippine EJournal. The study focused on improving operational efficiency in a small automotive repair shop by addressing the limitations of manual and semi-digital inventory and service management practices.

The researchers began by identifying the common challenges experienced in automotive workshop environments, particularly those that still rely on traditional tools such as Microsoft Excel, handwritten logs, or separate standalone files for managing operations. These fragmented systems often result in disorganized data handling, inconsistent updates, and difficulty in tracking the real-time status of inventory and ongoing service jobs.

One of the key issues highlighted in the study was the lack of system integration between different operational processes. In many repair shops, inventory tracking, customer records, and service transactions are handled separately, which means that information does not automatically flow from one process to another. For example, when a part is used in a repair job, it must be manually deducted from the inventory record. This manual dependency increases the risk of human error, such as forgotten updates, incorrect quantity entries, or duplicate records.

The study further emphasized that these manual processes become even more problematic in busy workshop environments where staff members are often multitasking between customer service, diagnostics, and repair work. Because of this workload, administrative tasks like updating inventory records are sometimes delayed or overlooked entirely. Over time, these small inconsistencies accumulate, leading to inaccurate stock levels and unreliable data.

Roque et al. (2023) also found that the absence of real-time data access significantly affects decision-making in automotive repair shops. Without updated inventory information, shop managers are unable to quickly determine whether a required part is available or needs to be reordered. This often leads to delays in repair services, especially when parts that were assumed to be in stock are actually unavailable. In some cases, this results in emergency purchasing from suppliers, which increases operational costs.

To address these challenges, the researchers developed a web-based Inventory and Service Management System using Agile methodology. The system was designed to centralize all key operational processes into a single platform, including inventory management, service tracking, customer records, and transaction processing. This integration ensured that all data within the system was interconnected and updated in real time.

A major feature of the developed system was automated inventory updating. Every time a transaction was recorded—whether a sale, part usage, or service completion—the system automatically adjusted stock levels without requiring manual input. This significantly reduced the risk of human error and ensured that inventory data remained accurate and up to date at all times.

Another important feature of the system was multi-user accessibility, which allowed different staff members to access and update the system simultaneously without causing data conflicts. This was particularly important in workshop environments where multiple employees may be working on different tasks at the same time. The centralized database ensured that all users were viewing and working with the same real-time information.

The study also incorporated Agile development principles, which involved continuous feedback from actual users throughout the system development process. This approach allowed the researchers to refine the system based on real operational needs rather than assumptions. As a result, the final system was more practical, user-friendly, and aligned with the daily workflow of automotive repair staff.

Testing and evaluation results from the study showed that the implemented system significantly improved operational efficiency. Users reported easier access to inventory data, faster transaction processing, and improved accuracy in stock monitoring. The reduction in manual workload also allowed staff to focus more on core repair and customer service activities rather than administrative tasks.

In relation to D Speedway, this study is highly relevant because it directly reflects the same operational environment and challenges faced by the shop. Like the workshop in Roque et al.'s study, D Speedway currently relies on Excel-based inventory tracking, which creates risks of inconsistent records, delayed updates, and manual encoding errors.

The concept of system centralization presented in the study is especially important for D Speedway's proposed Smart Inventory and Parts Planning System. By integrating inventory tracking, sales recording, and supplier management into a single platform, the system eliminates the need for multiple disconnected spreadsheets. This ensures that all data is consistent, updated, and accessible in real time.

Furthermore, the automated inventory updating feature developed in the study directly supports one of the core objectives of the proposed system for D Speedway. Every time a part is used or sold, the system automatically updates the stock count, reducing reliance on human memory or manual entry. This greatly minimizes errors and ensures that inventory records always reflect actual stock levels.

Overall, Roque et al. (2023) provides strong evidence that a centralized, automated inventory and service management system significantly improves efficiency, accuracy, and workflow

coordination in automotive repair environments. The findings strongly support the development of a Smart Inventory and Parts Planning System for D Speedway, as both systems aim to solve the same fundamental problems of manual tracking, data inconsistency, and lack of real-time inventory visibility.

Lirios et al. (2018), in their study published in Innovatus titled A Problem-Oriented Approach to Implementing an Auto Sales and Inventory System Through Agile Methodology, examined the operational challenges faced by an auto supply company that was heavily dependent on manual and semi-digital inventory practices. The study focused on identifying the root causes of inefficiencies in inventory tracking and sales recording, particularly in environments where transactions occur frequently and require timely updates.

The researchers found that the company's reliance on spreadsheet-based systems created persistent issues in data accuracy and consistency. Since encoding was done manually, inventory records were often updated late or incorrectly, especially during peak business hours when multiple transactions were happening simultaneously. This resulted in discrepancies between actual stock levels and recorded data, making it difficult for the business to maintain a reliable overview of available inventory.

The study further emphasized that the absence of a centralized system contributed significantly to operational inefficiency. Sales records, inventory updates, and product tracking were stored in separate files that were not automatically synchronized. Because of this, even small transactions required multiple manual updates across different documents. Over time, this fragmented structure increased the likelihood of missing entries, duplicated records, and mismatched stock information.

Another key issue identified was the difficulty in generating accurate reports for decision-making. Management had to manually consolidate data from multiple sources, which was time-consuming and prone to human error. As a result, decisions related to restocking, procurement, and sales forecasting were often based on incomplete or outdated information. This limitation directly affected the company's ability to respond quickly to changes in demand and supply conditions.

To address these challenges, Lirios et al. developed an automated inventory and sales management system using the Agile methodology. The system was designed to centralize all operational data into a single platform, ensuring that every transaction automatically updates both sales and inventory records in real time. This removed the need for manual cross-referencing between multiple files and significantly reduced the risk of inconsistencies in data entry.

The system also introduced structured workflows for recording transactions, monitoring stock levels, and accessing business reports. These features allowed employees to perform daily tasks more efficiently while ensuring that management had immediate access to accurate and updated information. The integration of automation also reduced dependency on individual memory and manual discipline, which are common sources of error in traditional systems.

In addition, the Agile development approach played a crucial role in the success of the system. By involving end users throughout the development process, the researchers were able to refine the system based on actual operational needs rather than assumptions. This iterative process ensured that the final system aligned closely with how the business operated in real-world conditions, making it more practical and easier to adopt.

The findings of the study showed that implementing an automated inventory system significantly improved data accuracy, reduced processing time, and enhanced overall workflow efficiency. It also strengthened the reliability of inventory records, which in turn improved the quality of decision-making within the company. The reduction in manual workload allowed employees to focus more on operational tasks rather than administrative corrections.

This study is highly relevant to the proposed Smart Inventory and Parts Planning System for D Speedway Automotive Repair Shop. Similar to the company in Lirios et al. (2018), D Speedway currently relies on Excel-based spreadsheets to manage sales and inventory data. This creates the same structural weaknesses identified in the study, including delayed updates, inconsistent records, and reliance on manual encoding for every transaction.

Both settings also share a common operational environment within the automotive industry, where inventory accuracy is critical to maintaining smooth repair workflows. Inaccurate or outdated stock records can lead to service delays, incomplete repairs, and customer dissatisfaction. The problems identified in the study closely mirror the challenges faced by D Speedway, particularly in ensuring that parts availability is accurately reflected at all times.

By applying the principles highlighted in the study—centralized data management, automation of transactions, and real-time inventory synchronization—the proposed system aims to eliminate the inefficiencies caused by manual spreadsheet-based tracking. This includes reducing human error, improving reporting accuracy, and ensuring that stock levels are always updated immediately after each transaction.

Furthermore, the study provides a strong justification for implementing a system that is both automated and user-centered. Just as Lirios et al. demonstrated the importance of aligning system design with actual user workflows, the proposed system for D Speedway is designed to fit naturally into existing shop operations without requiring major changes in staff behavior.

Overall, Lirios et al. (2018) establishes that manual and fragmented inventory systems are no longer sufficient for businesses dealing with frequent transactions and fast-moving stock. Their findings strongly support the transition toward automated, centralized inventory systems, making it a solid theoretical and practical foundation for the development of a Smart Inventory and Parts Planning System for D Speedway Automotive Repair Shop.

Sison, Oreiro, Camalit, and Ng (2019) conducted a study published in Innovatus titled A Problem-Oriented Approach to Implementing an Inventory and Point-of-Sale System for Company KCP. The study focused on the operational challenges experienced by a business that relied heavily on manual and semi-digital methods for handling sales transactions and inventory records. The researchers examined how the company's existing practices affected the overall efficiency of business operations, particularly in relation to transaction processing, stock monitoring, data organization, and reporting accuracy. Their work centered on understanding the actual problems encountered in the workplace and developing a system that directly addressed those operational weaknesses rather than simply introducing technology for the sake of modernization.

The researchers found that the company's existing Point-of-Sale and inventory process depended too much on manual data entry and disconnected record-keeping methods. Employees had to encode transactions separately, and stock updates were handled manually after sales were completed. Since the sales process and inventory monitoring process were not synchronized, the system became highly vulnerable to human error. During busy business hours, when multiple transactions were occurring at the same time, employees often struggled to keep records updated consistently. This resulted in delayed updates, missing entries, and inaccurate inventory counts that no longer reflected the actual stock available in the business.

The study emphasized that the problem was not simply the use of spreadsheets or digital files themselves, but the absence of integration between the different records being maintained. Sales records, inventory lists, refunds, and supplier information existed separately and required employees to manually compare files whenever verification was needed. This fragmented structure created confusion and slowed down normal operations because employees could not immediately determine which records were accurate and current. The researchers explained that once inconsistencies started appearing across files, staff members gradually lost confidence in the reliability of the records, forcing them to rely more on physical checking and manual confirmation instead of trusting the system itself.

Another major finding of the study was the issue of data redundancy and reduced data integrity caused by repetitive and disconnected encoding practices. Since employees needed to input the same information into multiple files or sections of the system, the possibility of duplicate entries and conflicting data became very common. For example, a transaction might be recorded correctly in the sales report but not reflected in the inventory file, or an inventory adjustment

might accidentally be entered more than once during reconciliation. Over time, these small inconsistencies accumulated and made reports increasingly unreliable for management use.

The study pointed out that unreliable data does not only create administrative inconvenience but also affects the quality of business decision-making. Managers depend on accurate reports to determine which products are selling quickly, which items should be reordered, and how much inventory should be maintained. When reports are based on inconsistent records, purchasing decisions become less accurate and less efficient. In some cases, businesses may reorder products unnecessarily, while in other cases they may fail to restock important items on time because the system incorrectly shows that inventory is still available.

The researchers also highlighted how inefficient inventory monitoring directly contributed to operational delays within the company. Since stock levels were not updated in real time, employees often discovered inventory shortages only when processing customer transactions. This reactive approach created disruptions in workflow because customer requests could not always be fulfilled immediately. Employees then had to spend additional time contacting suppliers, verifying stock availability, or manually checking inventory storage areas to confirm whether items were actually available.

The consequences of these delays extended beyond internal operations and affected customer experience as well. Customers who expected quick service were forced to wait longer because employees first had to resolve inventory issues before completing transactions. The study explained that repeated delays and stock-related interruptions could negatively influence customer satisfaction and reduce confidence in the business. At the same time, the company also experienced overstocking of slow-moving items due to the lack of accurate consumption tracking

and forecasting. Without reliable records showing actual product movement, inventory decisions were based largely on estimation rather than verified data.

To solve these problems, Sison et al. developed an automated Inventory and Point-of-Sale system designed to centralize all major business operations into a single platform. The system integrated sales processing and inventory monitoring so that every completed transaction automatically updated the corresponding stock level in real time. This removed the need for separate manual inventory adjustments and significantly reduced the possibility of inconsistent records across different files.

The proposed system also introduced structured database management features that improved the organization and reliability of information stored within the business. Instead of relying on scattered spreadsheets and independently maintained documents, the system stored all records within a centralized database environment where transactions, inventory data, and reports were interconnected. This structure ensured that all information remained synchronized and consistent throughout the system.

Another important contribution of the study was its focus on improving transaction speed and operational efficiency. Because employees no longer needed to switch between multiple spreadsheets or manually verify records during transactions, the processing time for customer purchases became significantly faster. Routine administrative tasks that previously consumed large amounts of time were simplified through automation, allowing employees to focus more on customer service and operational responsibilities rather than repetitive paperwork and data correction.

The researchers further emphasized the importance of usability and practicality when developing information systems for real business environments. They recognized that not all employees possess advanced technical skills, especially in small and medium-sized businesses where workers are more experienced in operational tasks than in information technology. Because of this, the system was designed with simplicity and accessibility in mind. Menus, transaction forms, and inventory functions were organized in a way that employees could understand and use with minimal training.

This user-centered approach became one of the strengths of the study because it addressed a common issue in system implementation: resistance to adoption. The researchers explained that even technically advanced systems often fail when users find them difficult, confusing, or disruptive to their normal workflow. By designing the system around the actual needs and routines of employees, the study increased the likelihood that the technology would be accepted and consistently used within the organization.

The results of the study showed that implementing a centralized Inventory and Point-of-Sale system significantly improved operational efficiency within the company. Data accuracy improved because transactions and inventory updates were synchronized automatically. Processing time decreased because employees no longer needed to manually compare multiple files or repeatedly encode the same information. Inventory monitoring became more reliable, allowing management to maintain better control over stock levels and reduce both shortages and unnecessary overstocking.

The study also demonstrated that automation can strengthen decision-making within a business environment. Because reports were generated from organized and continuously updated records,

management gained quicker access to accurate information regarding sales performance, inventory movement, and purchasing needs. This allowed the company to make more informed operational decisions based on actual data rather than assumptions or delayed reports.

This study is highly relevant to the proposed Smart Inventory and Parts Planning System for D Speedway Automotive Repair Shop. Similar to the company examined by Sison et al. (2019), D Speedway currently relies on manual and spreadsheet-based methods for handling sales and inventory management. The shop experiences many of the same operational difficulties identified in the study, including inconsistent inventory updates, disconnected records, delayed reporting, and the absence of real-time stock visibility.

Both environments also involve transaction-heavy operations where products and inventory move constantly throughout the day. In automotive repair shops such as D Speedway, inventory management is especially important because repairs depend on the immediate availability of specific automotive parts. Even small inaccuracies in stock monitoring can interrupt repair work, delay customer service, and create unnecessary operational stress for both mechanics and staff members.

The findings of the study clearly demonstrate that the challenges experienced at D Speedway are not isolated problems but are common among businesses that continue to rely on manual record management systems. The issues identified by Sison et al. — fragmented records, delayed inventory updates, inconsistent data, and inefficient reporting — closely reflect the current operational situation observed at D Speedway during the researchers' interviews and site visits.

By applying the principles presented in the study, particularly centralized data management, automation of transactions, real-time inventory synchronization, and simplified reporting, the proposed Smart Inventory and Parts Planning System aims to eliminate many of the inefficiencies currently affecting the shop. The proposed system will allow inventory levels to update automatically after every transaction, reduce dependence on repetitive manual encoding, and provide staff with a more reliable way of monitoring stock availability.

In addition, the study supports the importance of designing systems that fit naturally into the workflow of the business rather than forcing employees to completely change how they operate. This principle is highly important for D Speedway because the proposed system is intended not only to improve technical operations but also to remain simple, understandable, and practical for daily use by shop personnel.

Overall, the study conducted by Sison et al. (2019) provides strong support for the development of an integrated Smart Inventory and Parts Planning System. It demonstrates how centralized and automated inventory management can improve accuracy, strengthen operational efficiency, reduce administrative workload, and support better business decision-making. The study therefore serves as both a practical reference and a strong conceptual foundation for the proposed system to be developed for D Speedway Automotive Repair Shop

2.3 Related Literature

Several foundational concepts from business and software development literature directly shape the design and objectives of the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop. These are not abstract theories — they are practical principles that explain why the problems currently experienced at D Speedway exist, why manual systems eventually fail to keep up with the demands of a growing business, and why the features being built into the proposed system are designed the way they are.

Inventory Management.

Inventory management is one of the most critical operational functions in any business that depends on physical stock to deliver its services. The core principle is straightforward: businesses must avoid two failure modes at the same time. The first is overstocking, where too many parts are ordered and stored, tying up money that could be used elsewhere in the business and creating the risk that slow-moving items will become obsolete before they are ever used. The second is understocking, where too few parts are on hand to meet demand, leading to delays, incomplete repairs, and customers who are forced to wait or take their business elsewhere. Both outcomes carry real costs, and in automotive repair environments, where customers expect their vehicles to be serviced within a predictable timeframe, the consequences of either extreme show up quickly in the form of lost revenue and damaged reputation.

What makes inventory management genuinely difficult is that the right balance is not a fixed number — it shifts constantly based on how quickly parts are being used, what new vehicles are

coming into the shop, what seasonal patterns exist in demand, and how reliably suppliers can deliver when orders are placed. A business relying on manual records and personal memory to track all of this is essentially asking one or two people to maintain a running calculation across dozens or hundreds of individual parts, update it after every transaction, and catch any developing shortage before it disrupts operations. This is not a sustainable approach. It works at very small scale and for a short period of time, but as the volume of transactions increases and the variety of parts grows, manual tracking becomes a source of error rather than a source of control. The proposed system directly addresses this by automating the tracking process, removing the dependency on individual memory, and making the current state of inventory visible at any moment without requiring anyone to calculate or compile it manually.

Real-Time Inventory Monitoring.

One of the most consistent findings across inventory management literature is that the timing of information matters as much as its accuracy. A record that accurately reflects stock levels from three days ago is not the same as a record that reflects stock levels right now, and in a busy automotive repair shop where multiple repairs are happening throughout the day and parts are being pulled from storage constantly, that gap between what the record says and what is actually on the shelf can grow very quickly. When inventory records are updated manually at the end of the day, or only when someone remembers to do it, there is always a window of time during which the numbers in the system do not match the physical reality of the storage area. During that window, a mechanic might be sent to retrieve a part that the records show as available but that was actually used two hours earlier and never logged. The repair stops, time is wasted, and the customer waits.

Real-time inventory monitoring eliminates that window by connecting every transaction directly to the inventory count. When a part is recorded as used or sold, the system immediately adjusts the corresponding stock level without any separate manual step. This means the records are always current, and anyone checking the system at any point during the day is seeing an accurate picture of what is actually available. For D Speedway, where the research team observed that inventory shortages are sometimes discovered only after a repair is already underway, real-time monitoring is not a luxury feature — it is the core of what makes the proposed system more reliable than the spreadsheet-based approach currently in use. The difference between finding out a part is gone before the repair starts versus finding out in the middle of it is the difference between a managed situation and a disrupted one.

Sales Reporting and Business Decision-Making.

Accurate and accessible sales data is one of the foundations of sound business decision-making, and the speed at which that data can be retrieved is directly connected to how useful it actually is. When a business owner has to manually pull figures from multiple files, reconcile inconsistencies between records, and compile everything into a usable format before any analysis can begin, the reporting process itself becomes a barrier. It takes time that could be spent on other things, it introduces the possibility of calculation errors during compilation, and it produces a result that reflects the state of the business several hours or days ago rather than right now. For a small automotive repair shop where purchasing decisions, staffing adjustments, and service priorities all depend on knowing which parts are moving and how fast, a reporting process that is this slow and this labor-intensive is a genuine operational problem.

The proposed system addresses this by generating sales summaries and inventory reports automatically from the data that is already being recorded through normal transactions. No separate compilation step is needed. The owner of D Speedway should be able to check how a particular part has been selling over the past week, or pull the total value of parts used during the current month, in a matter of seconds rather than hours. This changes the relationship between the business owner and the data. When reports are easy to access, they get consulted more often. When they get consulted more often, purchasing decisions are grounded in actual patterns rather than estimates. When purchasing decisions are grounded in actual patterns, the business is better positioned to maintain the right stock levels, avoid unnecessary expenses, and respond quickly when demand shifts.

Reorder Point and Demand Forecasting.

A reorder point is the specific stock level at which a new order for a particular part should be placed, calculated based on how quickly that part is being consumed and how long it takes for a new delivery to arrive. The logic behind it is simple but powerful: rather than waiting until a part completely runs out before placing an order — by which point it is already too late to avoid a disruption — the business sets a threshold in advance and places the order while there is still enough stock to keep operating through the lead time. Done correctly, this means the shop never actually runs out of a part during normal operations. The new stock arrives before the existing stock is fully depleted, and the workflow continues without interruption.

Applying this in practice requires two things: accurate data about how fast each part is being used, and a system that monitors stock levels continuously and raises an alert when the reorder threshold is reached. Neither of these is possible without a properly functioning inventory management system. If the usage data is unreliable because records are updated inconsistently, the reorder point calculation will be off, and the alert will either fire too early — prompting unnecessary orders — or too late — after the shortage has already begun. If the stock levels are not monitored in real time, the alert may not fire at all until the next time someone manually checks the records, by which point the part may already be gone. The proposed system for D Speedway combines accurate real-time stock tracking with automated low-stock alerts and usage-based forecasting, giving the shop the tools to stay ahead of shortages rather than reacting to them after the fact. Over time, as the system accumulates more transaction history, the forecasting becomes more precise because it is drawing on a longer and more complete record of actual demand patterns.

System Design and Usability.

A system that is technically well-built but difficult to use in practice will not deliver the results it was designed to produce. This is one of the most consistent findings in software development and information systems literature, and it is particularly relevant in environments like automotive repair shops, where the people who will be using the system every day are not technology specialists. They are mechanics, service staff, and shop managers whose primary focus is on vehicles and customers, not on navigating software interfaces. If a system requires complicated data entry, a large number of steps to complete a basic task, or a significant adjustment in how staff members go about their normal routine, the likelihood that it will be used consistently and

correctly drops considerably. Systems that are resisted or used incorrectly do not improve the operational problems they were built to solve — they just add a new layer of complexity on top of the existing ones.

Good system design in this context means building software around the actual workflow of the business rather than asking the business to adapt to the software. It means making the most common tasks — recording a transaction, checking stock, reviewing a sales summary — as simple and fast as possible. It means reducing the number of steps required to complete each action, presenting information in a format that is immediately understandable without interpretation, and minimizing the chance that a user will make an error during normal use. The decision to observe D Speedway's daily operations before any design decisions were made, and to involve staff members in testing before the system is finalized, reflects this principle directly. The proposed system is being shaped by how the shop actually works, not by assumptions about how an inventory system should function in theory.

Web-Based Systems and Multi-User Accessibility.

Building the proposed system as a web-based platform rather than a desktop application is a practical decision grounded in how small automotive businesses actually operate. In a repair shop environment, multiple people interact with inventory and sales data throughout the day — mechanics pulling parts, counter staff recording sales, and the owner reviewing stock levels and reports. If the system lives on a single desktop computer, access becomes a bottleneck. Only one person can use it at a time, and anyone who is not physically at that machine cannot check or update the records without walking over to it. This creates delays and increases the likelihood

that updates will be deferred until someone is at the right machine, which is exactly the kind of inconsistency that leads to inaccurate records.

A web-based system removes this constraint entirely. Because it runs through a browser rather than a locally installed program, it can be accessed from any device on the network — a desktop at the front counter, a tablet in the storage area, or a phone in the workshop — without any additional setup. All users are interacting with the same centralized database at the same time, which means that when one person records a transaction, everyone else immediately sees the updated stock level. There is no version conflict, no duplicate file, and no risk of one person's update overwriting another's. For D Speedway, where the owner and staff are often in different parts of the shop and need current information at different points in the day, this kind of shared real-time access is not a minor convenience — it is a fundamental requirement for the system to work as intended. A centralized web-based platform ensures that the inventory records the owner is looking at are the same records the mechanics are working from, at all times.

2.4 Synthesis of the Reviewed Literature

Looking at everything reviewed in this chapter together, one thing becomes hard to ignore: the problems D Speedway is dealing with right now are not unique to this shop. Every study covered in this chapter, regardless of where it was conducted or what kind of automotive business it examined, kept running into the same story. A small business starts out managing its inventory manually — spreadsheets, handwritten logs, or a combination of both. It works well enough at first, when the number of transactions is low and the variety of parts is still manageable. But as the business grows and the volume of daily activity increases, the manual approach starts to

crack. Records fall behind. Stock counts become unreliable. The owner makes purchasing decisions based on memory rather than data. Parts run out without warning. Reports take hours to put together and still do not tell the full story. These are not the kinds of problems that happen because of poor management or careless employees — they happen because manual systems have a ceiling, and once a business grows past that ceiling, the system starts working against it instead of for it.

What the reviewed literature establishes very clearly is that this pattern repeats itself across different types of automotive businesses and different scales of operation. Whether the business is a spare parts store, a motorcycle shop, or a general repair shop, the moment it starts handling a volume of transactions that a spreadsheet cannot comfortably keep up with, the same set of problems appears. Inventory records drift away from reality because updates depend on people remembering to do them consistently, and in a busy shop environment, inventory updates are almost always the first thing that gets skipped when things get hectic. Stock shortages are discovered only after they have already caused a problem — a mechanic is already mid-repair when someone realizes the needed part is gone, or a customer is already waiting when staff discovers the part they ordered was actually out of stock. Reports that should inform daily decisions instead get compiled at the end of the week or the end of the month, by which point whatever information they contain is too late to act on. The businesses in the reviewed literature are not struggling because they are doing something wrong. They are struggling because the tools they are using were not built for the kind of work being asked of them.

The foreign studies covered in this chapter are particularly useful because they go beyond identifying the problem and actually document what happens when a proper digital system is put in place. Across every case where an automated inventory management system was developed and implemented, the results followed the same pattern: inventory accuracy improved, stock-related disruptions decreased, the time staff spent on manual administrative tasks dropped significantly, and the owner gained access to reliable business information without having to compile it by hand. These are not small improvements — they represent a fundamental change in how the business operates on a daily basis. When inventory records update automatically after every transaction, the owner stops guessing and starts knowing. When low-stock alerts fire before a part runs out rather than after, the shop stops reacting to shortages and starts preventing them. When sales reports are generated automatically from stored data, business decisions are made from facts rather than estimates. The literature shows, consistently and without exception, that these outcomes are achievable for small businesses of exactly the kind and scale that D Speedway represents.

The local studies add something important to this picture that the foreign research alone cannot provide. The challenges documented in automotive businesses closer to home confirm that the same operational problems experienced at D Speedway are not just a general business phenomenon — they are present specifically in small automotive shops operating under conditions that closely mirror D Speedway's own situation. Small staff, growing transaction volume, inventory managed through spreadsheets, no centralized system, purchasing decisions driven by personal judgment rather than tracked data. The businesses described in the local studies are dealing with exactly this, and the findings make clear that the problems are structural

rather than circumstantial. They do not go away on their own as the business matures. If anything, they compound. The longer a business runs on a manual system past the point where that system can handle the load, the more embedded the habits become and the harder the eventual transition gets. The local studies do not just confirm that the problem exists — they confirm that it exists in the same context as D Speedway, which makes their findings directly applicable to this project.

Taken together, the reviewed literature points to four things that any effective inventory system for an automotive repair shop must be able to do. First, it must update inventory records automatically every time a transaction occurs, without requiring any separate manual step. The moment human memory or human consistency becomes a dependency in the update process, errors become inevitable. Second, it must make current stock levels visible in real time so that the gap between what the records say and what is physically available is eliminated rather than just reduced. Third, it must be capable of tracking usage patterns over time and using that data to anticipate when parts will need to be reordered, so the shop can act on that information before a shortage occurs rather than after. Fourth, it must be simple enough to use reliably in a busy shop environment where the people operating it are not technology specialists and cannot afford to spend significant time learning complicated software. These are not wish-list features — they are the minimum set of capabilities that the literature consistently shows are necessary for an inventory system to actually solve the problems it is supposed to solve.

One more thing that comes through clearly across the reviewed studies is how much the design process matters, not just the design itself. The systems that worked well — the ones that got

adopted by staff and continued to be used consistently after implementation — were all built through a process that started with understanding the actual business before making any technical decisions. The developers visited the businesses in person, observed how daily operations worked, talked to the people who would be using the system, and let those observations shape the design. The systems that were built purely on technical assumptions without this kind of grounding tended to produce tools that staff found confusing or that did not fit naturally into how the work actually got done. For a shop like D Speedway, where the goal is a system that mechanics and counter staff can use confidently in the middle of a busy workday without it becoming another thing to worry about, this principle is not just worth noting — it is central to how this project is being approached.

What does not yet exist in the literature is a system that brings all of these elements together in a single integrated tool built specifically for a small automotive repair shop of D Speedway's profile. The reviewed studies show pieces of the solution working in various settings — real-time tracking here, demand forecasting there, automated reporting somewhere else, usability-centered design in another. Each of those contributions is valuable and each one informs a part of what is being built for this project. But none of them was built for this shop, and none of them addresses the specific combination of needs that D Speedway has: a young and growing repair business, operating with limited staff, handling a wide variety of vehicle types, managing a parts inventory that is already becoming too complex for spreadsheets, and needing a system that is both capable enough to handle that complexity and simple enough to be used by people whose expertise is in fixing vehicles, not managing software. That is the gap this project is designed to fill. The proposed Smart Inventory and Parts Planning System for D Speedway is not adapted

from something built for a different context. It is being developed from the ground up, shaped by the consistent findings of the reviewed literature and by direct knowledge of how this specific shop operates day to day. The literature tells us what the system needs to accomplish. The actual conditions at D Speedway tell us how it needs to accomplish them.

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CHAPTER 3

METHODOLOGY

3.1 Research Design

This study will employ a descriptive-developmental research design to guide the development of the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop. This research design will be appropriate because the study aims not only to understand the current inventory management practices of the business but also to develop a system that will address the operational issues identified during the investigation. By combining description and development within a single framework, the study will establish a direct connection between the problems observed in the existing process and the solutions incorporated into the proposed system.

The descriptive phase of the study will focus on examining the current inventory and sales management procedures of D Speedway Auto Repair Shop. Before the development of the system begins, a thorough understanding of the shop's existing operations will be established. This phase will provide a clear picture of how inventory is monitored, how sales transactions are recorded, how stock levels are tracked, and how purchasing decisions are made. Since every business has its own workflow, practices, and challenges, understanding the specific environment of D Speedway will be essential in ensuring that the proposed system is relevant to its actual operational needs.

Information will be gathered through direct observation of daily activities within the shop and through discussions with the owner and staff members who are directly involved in inventory and sales-related tasks. Observing actual work processes will provide valuable insights into how operations are carried out in real situations rather than relying solely on existing records or assumptions. Attention will be given to the methods used for recording transactions, updating inventory records, monitoring available stock, and generating reports. In addition, discussions with personnel will help identify common difficulties encountered during daily operations, as well as the information needed to perform tasks more efficiently.

Through this process, the study will identify operational challenges that affect the efficiency and accuracy of inventory management. Existing practices may reveal issues such as delayed inventory updates, inaccurate stock records, difficulties in monitoring inventory levels, and the time-consuming nature of manually preparing reports. The study will also examine how these challenges impact decision-making, customer service, and overall business operations. Understanding these concerns will be important because they will serve as the basis for determining the features and functions that should be included in the proposed system.

The descriptive phase will play a significant role in the success of the project because it will establish a clear foundation for development. Rather than adopting features commonly found in generic inventory systems, the proposed solution will be designed according to the actual requirements of D Speedway Auto Repair Shop. This approach will help ensure that every feature included in the system has a practical purpose and directly addresses a problem identified during data gathering.

Following the completion of the descriptive phase, the developmental phase will begin. During this stage, the Smart Inventory and Parts Planning System will be designed and developed based on the information gathered from the business. The development process will focus on creating a system capable of improving inventory monitoring, simplifying record management, and providing accurate information that can support daily operations and decision-making.

The proposed system will include features intended to address the operational gaps identified during the descriptive phase. These features may include automatic inventory updates whenever a transaction is recorded, real-time monitoring of stock quantities, low-stock notifications, demand forecasting capabilities, reorder recommendations, and automated report generation. Each component will be developed with the objective of reducing manual work, minimizing human error, and improving the accessibility of inventory information.

An iterative approach will be followed throughout the developmental phase. Instead of completing the entire system in a single development cycle, individual components and functions will be continuously reviewed and refined as development progresses. This approach will allow improvements to be made whenever issues related to usability, functionality, or workflow compatibility are identified. Adjustments will be based on observations and feedback gathered during system demonstrations and testing activities.

As different features become available, opportunities will be provided for the owner and staff members to interact with the system and evaluate its functionality. Their observations and suggestions will contribute to the refinement of the system, ensuring that its interface and

workflow remain aligned with the actual practices of the shop. This process will help ensure that the final product supports existing operations rather than creating unnecessary complications for users.

After the development process has been completed, the system will undergo testing and evaluation within the actual operating environment of D Speedway Auto Repair Shop. Conducting the evaluation in the workplace will make it possible to assess the performance of the system under real operational conditions. Actual inventory transactions, stock monitoring activities, and reporting tasks will be carried out using the system to determine whether it functions effectively in supporting daily operations.

The evaluation phase will focus on determining whether the system improves inventory accuracy, enhances stock monitoring, reduces the time required to manage records, and provides useful information for inventory planning and purchasing decisions. The usability of the system will also be assessed to determine whether users can perform their tasks efficiently with minimal difficulty. Any limitations identified during testing will be documented and considered when assessing the overall effectiveness of the proposed system.

The descriptive-developmental research design will provide a structured process for achieving the objectives of this study. The descriptive phase will establish a detailed understanding of the current operational environment, while the developmental phase will transform those findings into a practical technological solution. By maintaining a strong relationship between the

identification of problems and the development of system features, the study will ensure that the proposed Smart Inventory and Parts Planning System addresses the actual needs of D Speedway Auto Repair Shop. Through this approach, the system is expected to contribute to more efficient inventory management, improved planning of automotive parts, more accurate record-keeping, and better overall operational performance.

3.2 System Development Methodology

The team will follow the Agile methodology throughout the development process. The reason for choosing Agile over something like Waterfall is simple: when you are building for a real business, requirements tend to shift once you start showing people what the system actually looks like. Agile accommodates that by building in short cycles. Each cycle produces a working piece of the system that can be tested and adjusted before moving forward.

The Waterfall model, by comparison, works by completing one phase entirely before moving on to the next. Planning is done first, then design, then development, then testing — in that strict order. The problem with applying that to a project like this is that D Speedway's staff may not fully know what they need from the system until they actually see a version of it running. Something that sounds good in a requirements document might turn out to be confusing in practice, or a feature that nobody thought to mention during planning might turn out to be essential once the team starts using the system day to day. Waterfall does not leave much room to go back and adjust once a phase is closed. Agile does, and that flexibility is exactly what this kind of project calls for.

Another reason Agile fits here is that it keeps the development grounded. Instead of building everything and then testing it all at once at the very end, Agile delivers working pieces of the system regularly throughout the process. That means problems get caught early, when they are still easy to fix, rather than at the end when the cost of going back is much higher. It also means D Speedway can start seeing what the system looks like and how it works before it is fully finished, which gives the team a chance to weigh in while there is still time to make changes. That kind of ongoing involvement from the people who will actually use the system tends to produce a much better end result than handing something over at the finish line and hoping it matches what was needed.

The development will move through the following phases:

Planning: Review of D Speedway's current setup and definition of what the system needs to accomplish.

This phase is about getting a clear picture of where the shop stands before anything gets built. That means sitting down with the owner and walking through how things are currently being handled — which spreadsheets are being used, who updates them, how often, and what kind of information is being tracked in each one. It also means looking at where the process breaks down: where entries get missed, where files conflict, where the owner is spending time doing things manually that the system should eventually handle automatically. The goal of this phase is not just to understand the problem in general terms but to map it out specifically enough that the team knows exactly what the system needs to do and in what order those things need to be built. Without a clear plan going in, even a well-designed development process tends to drift, and time

gets spent on things that are not actually the priority. Planning keeps the project focused from the start.

Requirements Gathering: Collection of specific information through observation and interviews — what data needs to be tracked, who will use the system, and what problems need to be solved first.

If the planning phase is about understanding the problem from a high level, requirements gathering is about getting into the details. This is where the team spends time at the shop — watching how work actually flows throughout the day, sitting with staff as they handle transactions and check inventory, and asking the kinds of questions that only come up when you are seeing the process in person rather than just hearing it described. Interviews will be conducted with the owner, the mechanics, and whoever is currently responsible for managing the spreadsheets, because each of those people has a different relationship with the data and a different set of needs from the system. What the owner needs to see in a report is not the same as what a mechanic needs to know when checking stock, and what the staff member updating inventory needs is different again. All of those perspectives have to be captured and accounted for before the design work begins, because trying to add them in later almost always means redoing work that has already been done. The output of this phase is a clear, specific list of requirements — what the system must do, what it should do, and what is out of scope for this version.

System Design: Design of the database structure, interface layout, and system logic, keeping things simple enough for non-technical staff to use daily.

With the requirements in hand, this phase focuses on translating what the system needs to do into a concrete plan for how it will be built. The database structure has to be designed to hold all the relevant information — parts, stock levels, transactions, suppliers, sales history — in a way that is organized, efficient, and easy to query when reports are needed. The interface layout has to be thought through from the perspective of the people who will be using it every day, most of whom are not technical users. Buttons need to be where people expect them. Information needs to be displayed clearly without requiring any kind of interpretation. The fewer steps it takes to complete a common task, the better. System logic — the rules that govern how data flows and how the system responds to different actions — also gets defined here. When does a low-stock alert trigger? How does a sale automatically update the inventory count? What happens when the same part is received from a supplier and added back to stock? These are the kinds of questions that have to be answered at the design stage so that the development work that follows has a solid blueprint to build from rather than making decisions on the fly.

Development: Building the system feature by feature — starting with sales recording and inventory tracking, then adding reports, forecasting, and reorder tracking. Each feature is tested before the next is built.

This is the longest phase of the project and the one where the actual system starts to take shape. The decision to build feature by feature rather than all at once is a deliberate one. Starting with sales recording and inventory tracking makes sense because those are the most fundamental parts of the system — everything else depends on that data being captured correctly. Once those core features are working and tested, the team moves on to the next layer: generating reports that the

owner can actually use, building in the reorder alerts that flag low stock before it becomes a problem, and adding the forecasting tools that help the shop plan ahead based on historical patterns. Each feature goes through its own round of testing before development moves forward. That way, if something is not working the way it should, the problem gets caught immediately while the code is still fresh and the fix is straightforward. Building everything first and testing everything at the end is how you end up with a system full of interconnected problems that are hard to untangle. Testing as you go keeps the build clean and the progress steady.

Testing and Evaluation: Running the completed system through unit, integration, and user acceptance testing with real staff at D Speedway.

Testing happens in layers, and each layer serves a different purpose. Unit testing checks that each individual component of the system works correctly on its own — that a sale gets recorded properly, that the inventory count updates to the right number, that a report pulls the correct figures. Integration testing then checks that all of those individual components work correctly together, because something that functions fine on its own can still break when it interacts with another part of the system. These two rounds of testing are done by the development team and are focused on catching technical issues before the system is handed over to anyone outside the team.

User acceptance testing is different. This is where real staff at D Speedway — the people who will actually be using the system day to day — sit down with it and try to use it the way they would in a normal workday. They record sales, check stock levels, run reports, and navigate through the system the way they would if it were already deployed. The goal here is not just to

find bugs, though that is part of it. It is to find out whether the system actually works for the people it was built for — whether the interface makes sense to them, whether the flow of tasks feels natural, and whether there is anything they expected the system to do that it does not currently do. Feedback from this round of testing is taken seriously and folded back into the system before the final version is delivered. A system that the development team thinks is finished but that the actual users find confusing or incomplete is not a finished system. User acceptance testing is what closes that gap.

3.3 System Architecture

The Smart Inventory and Parts Planning System for D Speedway will be built on a three-layer architecture. This kind of structure is a standard and well-established approach for web-based information systems, and the reason it is used so widely is not because it looks good on a diagram — it is because it actually makes systems easier to build correctly, easier to fix when something goes wrong, and easier to update later without breaking everything else. When the visual design, the processing logic, and the data storage are kept separate from each other, a change in one layer will not automatically create problems in the other two. That separation matters a lot in a real working environment because needs change over time. A new report format, a new field in the inventory form, a new alert condition — any of these will be addressable without touching the parts of the system that are already working. For a small business like D Speedway that may want to expand or adjust the system as it grows, that flexibility will not be a minor technical detail. It will be what makes the system sustainable beyond just the first month of use.

Each of the three layers will have a distinct role, and together they will handle every interaction that happens when the owner or manager uses the system — from the moment they log in to the moment a sales report is generated and printed. The following sections describe each layer in detail, explain what it will be responsible for, and trace how data will move through the entire system during a typical transaction.

User Interface Layer

The User Interface Layer will be what the owner and manager actually see when they open the system on their browser. It will be the part of the system that the user interacts with directly — every button they click, every form they fill out, every number they read on a report will come from this layer. Because D Speedway's staff are automotive professionals and not IT specialists, this layer will be designed with one guiding principle: if someone with no technology background cannot figure out how to use it in a single session, the design will need to be fixed.

The first screen a user will encounter will be the login page. It will require a username and password, and the system will use these credentials to determine what that user is allowed to see and do. The owner will have full access to every part of the system — inventory records, sales history, reorder management, expense tracking, reports, and all dashboard data. Staff members who are assigned limited roles will see only the parts of the system relevant to their responsibilities, such as recording sales transactions and checking stock availability. This role-based access structure will exist for a practical reason: it will prevent someone from accidentally

editing or deleting records they had no business touching in the first place, which is a real risk in any shared digital system.

After logging in, the user will be taken directly to the main dashboard. The dashboard will be the operational center of the entire system, and it will be designed to give the owner a clear, current picture of the business without requiring any manual data gathering. It will display a set of key performance indicators that will update automatically as transactions are recorded throughout the day. These will include total daily revenue broken down into parts sales and labor fees, the monthly revenue trend showing how income has moved over the current year, the top ten best-selling parts within a selected period, a revenue split showing what proportion of income is coming from parts versus services, a color-coded inventory alert panel showing which parts have dropped to or below their minimum stock threshold, and a payment method breakdown showing the split between cash and GCash transactions for the month.

These dashboard metrics will not be decorative. Each one was chosen because it answers a specific question that the owner at D Speedway currently has to ask repeatedly under the manual system. Right now, knowing how much revenue came in today means opening a spreadsheet and manually adding up rows of data. Knowing which parts are running low means physically checking the storage area or hoping someone remembered to update the inventory file. Knowing the monthly revenue trend means pulling multiple files and cross-referencing them. The dashboard will replace all of that with a single screen that will have already done the work. The owner will be able to open the browser, look at the dashboard for two minutes, and know more

about the current state of the business than they could previously learn from an hour of manual file review.

Beyond the dashboard, the User Interface Layer will include several other sections that the owner and manager will use for day-to-day operations. The sales recording form will be where transactions are entered — the user will select the parts used or sold, enter the quantities, record the payment method, and the system will take it from there. The inventory section will display the current stock level of every item in the catalog, searchable by name or category, with visual indicators showing which items are in a healthy stock range, which are approaching the minimum threshold, and which have already fallen below it. The reorder management section will allow the manager to record a pending reorder for a supplier, track its status, and mark it as received when the stock arrives. The expense tracking section will capture operational costs outside of parts purchases, which will feed into the profit and loss calculations available in the reports section. The reports section will allow the owner to generate daily, monthly, and annual summaries of sales performance, inventory movement, and revenue — and to export or print them as needed.

All of these interface elements will be built using a design that keeps visual noise to a minimum. The color scheme will use red to indicate critical low-stock situations, yellow for approaching-threshold warnings, and green for healthy stock levels. This color coding will allow a staff member to scan the inventory page and immediately understand which items need attention without reading every row in detail. Text labels will be written in plain language rather than

technical jargon, and navigation between sections will be done through a consistent sidebar menu that will behave the same way throughout the system. The goal throughout the design process will be to make sure that the system feels familiar enough by the second time someone uses it that they will not need to think about how to operate it — they will be able to just use it.

Application Layer

The Application Layer will be the part of the system that the user never sees but will depend on entirely. It will be the engine running underneath the interface — the set of processes and logic that will receive input from the user, act on it, update the database, and send back the right information to display on the screen. Every time something happens in the User Interface Layer, the Application Layer will be what makes it actually do something meaningful.

When a sale is recorded, the Application Layer will not just save a record of the transaction and stop there. It will immediately execute a sequence of operations in the background. First, it will save the transaction details to the database — the date, the parts involved, the quantities, the prices, the payment method, and the user who recorded it. Second, it will subtract the correct quantities from the inventory records for every part included in that transaction. Third, it will check each of those parts against its minimum stock threshold. If a part's new stock level has fallen at or below the threshold, the Application Layer will flag that item as low stock and queue a visual alert to appear on the dashboard and the inventory page. Fourth, it will update the revenue totals and KPI calculations that feed the dashboard metrics so that the next time the dashboard is viewed, it will reflect the transaction that was just recorded. All of this will happen

within the same moment that the user clicks the save button — the system will not wait to be told to check stock levels or update the dashboard. It will do it automatically, every single time.

The Application Layer will also handle the forecasting calculations that will support the reorder recommendation feature. At defined intervals, it will analyze the transaction history stored in the database to calculate the average consumption rate for each part over a rolling period. It will then use that rate to estimate how many units of each part the shop is likely to need in the coming weeks based on the pace at which those parts have been used in recent history. These forecasting outputs will be stored and made available through the inventory and reorder sections of the interface, where the manager will be able to see not just current stock levels but a projected depletion timeline for each item. This will not require anyone to calculate anything manually — the Application Layer will run the analysis using the data that is already in the database from every recorded sale.

For the reorder recommendation feature specifically, the Application Layer will cross-reference the forecasted depletion timeline with the existing reorder records to determine which items do not yet have a pending reorder despite having a projected shortage. It will also pull the supplier information associated with each part from the database and make it available alongside the reorder suggestion, so the manager will not have to search through records to find out who to contact. The manager will see the part that needs restocking, the suggested quantity based on past purchasing patterns, and the supplier that has been used before — all in one place. They will

then make the decision and create the reorder record manually, which the Application Layer will track through a status system from pending through ordered to received

.

The login and session management logic will also live in the Application Layer. When a user submits their credentials on the login page, the Application Layer will check them against the database, confirm the role associated with that account, and establish a session that will control what the user can access throughout their time in the system. When the session ends — either because the user logs out or because the session times out after a period of inactivity — the Application Layer will clear it so that the next person to open the browser will not be able to access the previous user's session. This will be a basic but essential security measure for a system that will contain business-sensitive financial and inventory data.

The Application Layer will be built using PHP, which will handle all of the server-side logic described above. PHP was chosen because it is well-suited for web-based systems of this scale, it has extensive support and documentation available, and it works seamlessly with the MySQL database that will be used in the Database Layer. The communication between the Application Layer and the User Interface Layer will be handled through standard HTTP requests, which means the system will run in a normal web browser without requiring any special software to be installed on the user's device.

Database Layer

The Database Layer will be where all of the shop's data lives. Every part in the catalog, every recorded transaction, every user account, every reorder record, every expense entry, and every stock level will be stored here. It will be the foundation that both other layers depend on — the User Interface Layer will read from it to display information, and the Application Layer will write to it whenever something changes. Without an accurate and well-structured database, neither of the other layers will be able to do their jobs properly.

The database will be built in MySQL and organized into a set of related tables, each one designed to hold a specific type of information. The parts catalog table will store the details of every item the shop carries — the part name, category, unit, selling price, cost price, minimum stock threshold, and current quantity on hand. The transactions table will record every sale that is entered into the system, with each row linked to the specific parts involved, the quantities sold, the prices at the time of sale, the payment method, and the date and time the transaction was recorded. The transaction items table will handle the relationship between a single transaction and the multiple parts it may include, since a single repair job will often involve more than one part being used at the same time.

The reorder records table will track every restocking action that the manager initiates through the system. Each record will contain the part being reordered, the quantity requested, the supplier being contacted, the date the reorder was created, and the current status — whether it is pending, ordered, or already received. When a reorder is marked as received, the Application Layer will

update the corresponding inventory quantity in the parts catalog table, closing the loop from low-stock alert to restocked shelves within the same system. The supplier table will hold the contact information and details for each supplier the shop works with, and it will be linked to the parts catalog so that the system will always know which supplier to surface when a reorder recommendation is generated for a specific part.

The expenses table will record operational costs that are separate from parts purchasing — utilities, labor, and other overhead expenses that factor into the shop's overall financial picture. These records will be used by the Application Layer when generating the profit and loss summaries in the reports section, giving the owner a more complete view of financial performance than sales revenue alone would provide. The users table will store the login credentials and role assignments for every account in the system, and it will be what the Application Layer checks against during authentication. Each user record will contain a hashed password rather than the password itself, meaning even if someone gained access to the database directly, they will not be able to read the passwords stored there.

The structure of the database will be designed with two priorities in mind: speed and reliability. Fast lookups will matter because the dashboard will need to load quickly every time the owner opens it — if the database queries take too long to run, the system will become frustrating to use in a busy shop environment. The tables will be indexed on the columns that are queried most often, such as part names in the catalog, dates in the transaction history, and status values in the reorder records, which will keep query times low even as the volume of stored data grows over

time. Reliability will matter because the database will be the single source of truth for every business decision the owner makes using the system. Data that is stored incorrectly or that can be lost due to a system failure will be worse than no data at all, because it will create false confidence. The database structure reflected in the Entity Relationship Diagram in the Supporting Diagrams section was designed to prevent those kinds of structural errors from the ground up.

Flow of a Typical Transaction Through the System

To make the three-layer architecture concrete rather than abstract, it helps to trace what will actually happen from the moment the owner or manager opens the system to the moment a completed transaction has been recorded and the inventory has been updated. This flow represents the most common daily interaction with the system and the one that the entire architecture was designed to support efficiently.

The manager will open a web browser and navigate to the system's login page. They will enter their username and password and click the login button. The User Interface Layer will send those credentials to the Application Layer, which will check them against the users table in the database. If the credentials match and the account is active, the Application Layer will establish a session, retrieve the user's role, and tell the User Interface Layer to load the dashboard. The dashboard page will request the current KPI data — today's revenue, current low-stock alerts, recent transactions, monthly trend figures, and top-selling parts — from the Application Layer,

which will pull all of that from the database and return it. The dashboard will load and the manager will be able to see the current state of the shop at a glance before doing anything else.

The manager will navigate to the sales recording section. A form will load with fields for selecting the parts involved in the transaction, entering the quantities used, specifying the labor fee if applicable, and choosing the payment method. The manager will fill in the form based on the repair that was just completed. When they click save, the User Interface Layer will send all of that form data to the Application Layer. The Application Layer will validate the data first — checking that all required fields are filled and that the quantities entered are not greater than the current stock levels. If validation passes, it will begin the write sequence: it will save the transaction header record to the transactions table, save each individual part entry to the transaction items table, and then run an update query on the parts catalog table to subtract the correct quantity from the stock level of each part that was used.

Immediately after updating the stock levels, the Application Layer will run a threshold check. It will compare the new stock level of each affected part against its minimum stock threshold. If any part's new level is at or below its threshold, the Application Layer will mark that part as low stock in the database. The next time the dashboard or inventory page is loaded — which may be within seconds if another tab is already open — the color-coded alert for that part will appear automatically. The manager will not have to remember to check. The system will flag it and put it in front of them.

If the manager notices a low-stock alert and decides to act on it, they will navigate to the reorder management section. The system will display the flagged parts alongside their current stock levels, their minimum thresholds, the suggested reorder quantities calculated from purchasing history, and the supplier information linked to each part. The manager will review the suggestions, adjust the quantities if needed based on their own judgment, and create a reorder record. The Application Layer will save this record to the reorder records table with a status of pending. When the supplier delivers the stock and the manager marks the reorder as received, the Application Layer will add the received quantity back to the part's stock level in the parts catalog table and update the reorder record status to received. The alert will clear from the dashboard.

At any point during or after this process, the manager will be able to navigate to the reports section and generate a sales summary for the current day, the current month, or any selected period. The Application Layer will query the transaction history from the database, aggregate the data by date, part, category, or payment method depending on what the report requires, and return the formatted results to the User Interface Layer for display. If the manager wants to review the profit and loss for the month, the Application Layer will combine the sales revenue from the transactions table with the expense records from the expenses table and return the net figure. All of this will be available within seconds and without any manual compilation.

What this flow demonstrates is that the three-layer architecture will not just be a structural choice made for technical reasons — it will be the reason the system can do all of these things together,

reliably, in real time, for a business that currently has to do every single one of these steps by hand. The User Interface Layer will make the system accessible to the owner and staff without requiring technical knowledge. The Application Layer will handle all of the logic that would otherwise fall to human memory and discipline. The Database Layer will store everything accurately and make it retrievable instantly. Each layer will have a specific job, and together they will produce a system that fits into D Speedway's daily operations without asking anyone to work harder — only smarter.

3.4 Tools and Technologies to be Used

The following tools will be used to build the system:

Frontend: HTML, CSS, and JavaScript for building the interface that users will interact with.

Backend: PHP, given its compatibility with MySQL and the team's familiarity with it.

Forecasting: CHART.JS for the parts demand forecasting component.

Database: MySQL, which is already reflected in the ERD design.

Development Environment: Visual Studio Code & XAMPP

3.5 Data Gathering Techniques

In order to build a system that actually addresses the real needs of D Speedway, it will be important to first develop a thorough understanding of how the shop currently operates. This

means going beyond assumptions and making sure that every design decision made throughout the development process will be grounded in actual information gathered directly from the source. Two data gathering methods will be used for this purpose — observation and interviews. Each method will serve a different function, and together they will provide a well-rounded picture of the shop's existing processes, its pain points, and what the people working there genuinely need from a new system.

The reason these two methods were chosen over others — such as surveys or questionnaires — is that D Speedway is a small, single-branch automotive repair shop with a small team, and the kind of information needed to design this system well cannot be captured through a checklist. It requires being present in the environment and talking directly to the people who deal with the problems every day. Surveys are useful for collecting data from large numbers of respondents, but for a study like this one where the goal is to understand the specific, day-to-day operational reality of one particular shop, direct observation and face-to-face interviews will produce far more useful and accurate information. The combination of both methods also serves as a form of cross-verification: what is observed in practice can be compared against what the owner and staff report in interviews, and where those two sources agree, there will be strong confidence that the finding is accurate and worth designing around.

Observation

The first method that will be used is observation. Time will be spent visiting D Speedway and watching how daily operations are actually carried out on the ground. Rather than relying on second-hand descriptions or asking staff to explain their workflow verbally, things will be seen firsthand — which tends to reveal details that people would not necessarily think to mention if they were just answering questions. People who have been doing the same tasks the same way for a long time often stop noticing the small workarounds they have built into their routines. Direct observation is one of the most reliable ways to bring those kinds of details to the surface because it captures what is actually happening in practice, which is not always the same as what people believe is happening or what they would describe out loud.

The observation sessions will take place during actual working hours so that the shop can be seen under real operational conditions — not during a quiet period set aside specifically for the visit, but during the kind of busy, unscripted day that D Speedway's staff deal with regularly. This matters because the problems being investigated are problems that appear under pressure: when multiple jobs are running at the same time, when a customer is waiting at the counter, when someone needs to check a part but also needs to answer a phone. Those are the conditions under which manual systems show their weaknesses most clearly, and those are the conditions under which the new system will need to perform.

During the observation sessions, attention will be given to several specific aspects of how the shop currently runs. The following are the key areas that will be closely monitored and documented:

- Transaction Recording Process — How sales are currently being recorded will be examined in detail, including whether the method used is paper-based, spreadsheet-based, or something else entirely. The time it takes to complete a single transaction entry will be noted, along with any moments where the process slows down, becomes complicated, or gets skipped entirely because something else demands the staff member's attention at the same time.
- Stock Level Checking — How staff members currently determine whether a part is available will be observed. This includes whether they physically walk to the storage area to count items, consult a spreadsheet file, rely on memory, or use some combination of all three. The accuracy and consistency of each approach will be noted, as well as how much time these checks take away from other tasks.
- Inventory Update Workflow — The process by which inventory records are updated after a part is used or sold will be tracked. Attention will be paid to how quickly updates happen after a transaction occurs, whether updates are sometimes delayed until the end of a shift or forgotten entirely, and whether different staff members handle this task in consistent or inconsistent ways.
- Sales Report Generation — How the owner or manager currently compiles sales reports will be examined, including which files are opened, how data is extracted and combined, and how long the process takes from start to finish. Any patterns of inconsistency or confusion between different files will be noted.

- **Reordering and Stock Replenishment** — The process by which the shop decides to reorder parts will be observed. This includes how and when the decision is made, whether it is triggered by a physical discovery that something has run out or by a proactive check of records, who makes the decision, and how the order is placed and tracked afterward.
- **Overall Shop Pace and Workflow Patterns** — The general rhythm of the shop throughout the day will be observed, including how busy different periods are, how staff move between their primary automotive work and administrative tasks, and what kinds of interruptions or unexpected problems tend to disrupt normal operations. This broader picture will be important for understanding the environment in which the new system will need to operate.

All observations will be written down carefully in real time so that the details remain accurate and can be referred back to during the design and development stages. The notes taken during these sessions will not just describe what was seen but will also capture the sequence of events, the approximate time each task took, and any points where something clearly went wrong or required an extra step to correct. These records will serve as a direct reference for deciding which system features are most urgently needed and how those features should be designed to fit into the shop's actual workflow rather than an idealized version of it.

Interviews

The second method that will be used is interviews. After the observation phase has been completed, the shop owner and selected staff members will be sat down with directly and asked specific questions about their experience with how things are currently being handled and what they would want to see in something new. While observation gives a clear picture of how things are done from the outside, interviews give the people involved a chance to speak for themselves — to explain what frustrates them, what they find difficult to deal with on a daily basis, and what they believe could genuinely be improved about the way the shop manages its records and inventory.

The decision to conduct interviews after observation rather than before it is intentional. Going into the interview sessions with firsthand knowledge of how the shop actually operates will allow for more focused and meaningful questions. Instead of asking vague, open-ended questions about inventory management in general, the questions will be able to refer directly to what was observed — specific moments, specific tasks, specific situations — and ask the owner and staff to elaborate on those. This makes the interviews more productive because the conversation starts from a shared, concrete understanding of the operational reality rather than from scratch.

The interview participants will be selected based on their role and their level of direct involvement with the processes the system is being built to address. The shop owner will be interviewed because they are the primary decision-maker and the person who will ultimately be responsible for using the system's reporting, forecasting, and reorder features on a regular basis. Staff members who are involved in day-to-day transaction recording and inventory checking will also be included because they are the ones who experience the friction of the current system most

directly during working hours. Their input will be especially important for ensuring that the system's interface and workflow are designed in a way that actually fits how they do their jobs.

The questions prepared for these interviews will be focused and purposeful. The following are the main areas that the interview questions will cover:

- **Daily Pain Points and Time-Consuming Processes** — Staff will be asked to describe which parts of their current workflow take the most time and cause the most frustration. This will include questions about how long it takes to record a transaction, how often they have to stop what they are doing to check on stock, and how much of their day is spent on administrative tasks rather than on the automotive work they were hired to do.
- **Common Errors and Recurring Mistakes** — The owner and staff will be asked to describe the kinds of mistakes that happen most often in the current system. This includes miscounted stock, missing transaction entries, incorrect figures in reports, data entered in the wrong file, or situations where two people's records contradict each other. Understanding where errors tend to occur will directly inform how the system will be designed to prevent those same mistakes from happening again.
- **Stock Shortage Experiences** — Both the owner and staff will be asked about specific instances where a part ran out unexpectedly during a repair job, and what happened as a result. This will include questions about how often these situations occur, what the usual

response is when they happen, and what the cost of that response tends to be — both in terms of money paid for rush orders and time lost while the job is on hold.

- **Reporting Challenges** — The owner will be asked in detail about how they currently generate sales and inventory reports, how long the process takes, and how confident they are in the accuracy of the figures they end up with. Questions will also be asked about what kinds of business decisions depend on those reports and whether the current reporting process gives them the information they actually need to make those decisions well.
- **Feature Priorities and Desired Capabilities** — The owner and staff will be asked what they would most want a new system to be capable of doing. This is where a clearer sense will be gained of which features matter most to the people who will be working with the system on a daily basis. It would not make sense to spend development time building something that the staff does not find useful or practical, so gathering this kind of direct input early on is a step that cannot be skipped.
- **Concerns About Adoption and Usability** — The owner and staff will also be asked about their comfort level with technology and whether they have any concerns about learning or using a new system. This information will be used to guide the design of the interface, particularly in terms of how much complexity is appropriate and where simplicity should take priority over additional features.

The interviews will be conducted in a semi-structured format. This means that a prepared set of questions will be brought into each session, but the conversation will not be rigidly limited to those questions alone. If an answer opens up an important direction that was not anticipated, that thread will be followed. The goal of the interview is not to fill out a checklist — it is to understand the actual experience of the people running D Speedway, and that requires a format that allows the conversation to go where it naturally needs to go.

The responses gathered during each interview session will be written down carefully, and where the participant's permission is given, notes will be taken in as much detail as possible so that nothing important is lost. After all interviews have been completed, the responses will be reviewed alongside the observation notes collected during the first phase. Where both sources point to the same problem or the same need, that will be treated as a strong and reliable signal that the system needs to address it. Where the two sources contradict each other — for example, where the observed reality does not match what someone described in the interview — those discrepancies will be examined more closely to understand why the gap exists and which source is more representative of the actual situation.

Taken together, the observation and interview data will serve as the foundation for every design and development decision made throughout this project. The features built into the system, the way the interface is organized, the alerts that are triggered, the reports that are generated — all of these will trace back to something specific that was seen or heard at D Speedway during the data gathering phase. That connection between the system's design and the shop's actual operational

reality is what the research team believes will give the final product its practical value, not just as a technical output of the study but as something the owner and staff will genuinely use and benefit from every day.

3.6 Testing Procedures

Testing will not be treated as a single activity that happens only after the entire system has been built. Rather, it will be conducted continuously and in phases throughout the whole development process. This approach is being chosen because catching problems early — while a feature is still being worked on — is far less complicated than trying to trace an error back to its source after several other features have already been added on top of it. When a system is tested only at the end, errors can be deeply buried and difficult to isolate, and fixing them often means revisiting code that has already been integrated with other parts of the system — which risks creating new problems in the process of solving old ones. Testing at each stage prevents that chain of complications from developing in the first place.

Three distinct testing phases will be carried out over the course of the project: unit testing, integration testing, and User Acceptance Testing. Each phase serves a different purpose and looks at the system from a different angle. Unit testing will confirm that each individual feature works correctly on its own before being connected to the rest of the system. Integration testing will confirm that all of those individually working features behave correctly when they are made to interact with each other. User Acceptance Testing will confirm that the system as a whole is practical, understandable, and genuinely useful for the people at D Speedway who will be using

it every day. Together, these three phases will cover the system from the level of a single function all the way through to the full experience of a real user doing real work.

Unit Testing

The first phase that will be carried out is unit testing. As the name suggests, this type of testing focuses on one unit or one feature at a time, completely separate from everything else in the system. The idea behind this is straightforward — before anything is connected to anything else, each individual piece needs to work correctly on its own. No feature will be considered ready to move forward until it has been tested and confirmed to be functioning the way it is supposed to. This approach keeps development clean and organized because it means that when something does not behave correctly, there is no need to search through an entire integrated system to find the cause. The problem is isolated by default because only one thing is being looked at at a time.

Unit testing will be performed by the developers immediately after each feature is coded, before moving on to the next one. Each test will involve feeding the feature a set of inputs — both valid ones and deliberately invalid or edge-case ones — and verifying that the output matches what was expected. Testing only the straightforward cases is not enough. A feature also needs to be tested against unusual inputs, missing data, extreme quantities, and boundary conditions to make sure it does not produce unexpected behavior under the kinds of circumstances that come up in a real working environment. A stock update function that works correctly when one unit is sold but breaks when twenty are sold at once is not ready, even if it passed the basic test.

The following are the specific features and functions that will go through unit testing:

Sales Transaction Recording — The function that saves a sale to the database will be tested to confirm that all required fields are saved correctly, that the transaction is linked properly to the right parts and quantities, and that incomplete or incorrectly formatted inputs are rejected with an appropriate error message rather than saved as corrupted records.

Automatic Inventory Deduction — The function that subtracts quantities from the parts catalog after a sale is recorded will be tested to confirm that the correct quantity is deducted from the correct item, that the deduction applies to every part included in a multi-part transaction, and that the stock count does not go below zero even if the entered quantity exceeds what is currently available.

Minimum Stock Threshold Alert — The function that checks whether an item's updated stock level has dropped to or below its defined minimum threshold will be tested to confirm that the alert is triggered at the right level, that it is not triggered prematurely, and that it correctly identifies the specific item that needs attention rather than flagging unrelated parts.

Reorder Record Creation and Status Tracking — The function that creates a reorder record and manages its status will be tested to confirm that new reorder entries are saved with the correct part, quantity, supplier, and date, that the status can be updated from pending to ordered

to received without errors, and that marking a reorder as received correctly adds the restocked quantity back to the inventory count.

Sales Report Generation — The reporting function will be tested to confirm that it correctly pulls transaction data from the database for the selected time period, that the totals calculated are accurate, and that the report displays the breakdown of parts revenue and labor revenue correctly without mixing or omitting any figures.

Dashboard KPI Calculations — Each metric displayed on the dashboard — daily revenue, monthly trend, top-selling parts, stock alert panel, revenue split, and payment method breakdown — will be tested individually to confirm that the figures are being pulled and calculated correctly and that they update when new transactions are recorded.

User Login and Authentication — The login function will be tested to confirm that valid credentials grant access, that invalid credentials are rejected with an appropriate message, that access levels are correctly applied after login so that each user only sees what their role permits, and that session timeout works as intended.

Expense Recording — The function that saves an expense entry will be tested to confirm that the correct amount, category, date, and description are saved, and that the expense figures are correctly reflected in the profit and loss calculations in the reports section.

After each function passes its unit tests, a record of the test cases used and the results observed will be kept. This documentation will serve as a reference during integration testing and will make it easier to re-run specific tests if a later change to the codebase needs to be verified against earlier behavior.

Integration Testing

Once a number of features have been individually built and confirmed to be working on their own, the next phase that will take place is integration testing. At this point, the focus will shift away from individual features and toward how all of those features function when they are connected and running together as a single system. It is entirely possible for two features to work perfectly on their own but then behave unexpectedly when they are made to interact with each other, and integration testing is specifically designed to catch that kind of problem. The reason this happens is that individual features are developed with certain assumptions about what the other parts of the system will provide or expect, and sometimes those assumptions do not match up perfectly when everything is finally combined.

The main thing that will be verified during this phase is whether the system's different components are properly communicating and responding to each other in the right sequence. The system is built around a chain of connected operations, and the reliability of that chain as a whole

is what integration testing is designed to confirm. If any link in the chain breaks — if one component fails to pass information correctly to the next, or if two components interfere with each other in an unexpected way — the system will not behave reliably under real working conditions. Finding and fixing those breaks during integration testing is far better than discovering them during actual use at the shop.

The following are the key integration scenarios that will be tested during this phase:

Sale-to-Inventory Update Chain — When a sale is recorded through the sales form, the system should automatically subtract the correct quantities from the parts catalog in the database. This chain will be tested by recording transactions involving different combinations of parts and quantities and then verifying that the inventory counts in the database reflect exactly what was sold. Edge cases such as transactions involving the same part multiple times in a single sale will also be tested.

Inventory Update-to-Alert Trigger Chain — When an inventory update brings a part's stock level to or below its minimum threshold, the system should immediately flag that item and display a low-stock alert on the dashboard and inventory page. This chain will be tested by deliberately recording transactions that bring specific parts to and below their thresholds and verifying that the alerts appear correctly in both locations at the right moment.

Reorder Received-to-Inventory Restoration Chain — When a reorder record is marked as received, the system should add the restocked quantity back to the part's current inventory count

and clear the low-stock alert if the new level is above the threshold. This entire sequence will be tested from the creation of the reorder record through to the confirmation of receipt and the verification of the updated stock level.

Transaction Data-to-Dashboard KPI Update Chain — When a new transaction is recorded, the dashboard figures — particularly the daily revenue total, the top-selling parts ranking, and the payment method breakdown — should update to reflect the new data. This will be tested by recording multiple transactions in sequence and verifying that the dashboard reflects each one accurately and without delay.

Expense Entry-to-Profit and Loss Report Chain — When an expense is recorded, it should be correctly factored into the profit and loss summary in the reports section. This will be tested by recording expenses of various amounts and categories and then generating reports to verify that the net figures are accurate and that the breakdown between revenue and expenses is correctly displayed.

Login-to-Access Level Enforcement Chain — When a user logs in with a specific role, the system should restrict their access to only the sections and functions permitted for that role. This will be tested by logging in with accounts assigned to different roles and attempting to access

sections that should be restricted, verifying that the system correctly denies those attempts without errors.

Integration testing will be repeated each time a new feature is added to the system, not just once at the end. This means that every new addition will be checked against everything that is already in place, and any conflict or unexpected behavior that the new feature introduces will be caught immediately rather than being discovered later when the problem has become more difficult to trace. A log of each integration test — what was tested, what inputs were used, what was expected, and what actually happened — will be maintained throughout this phase so that the testing process is documented and repeatable.

User Acceptance Testing (UAT)

The final phase of testing will be User Acceptance Testing, which is more commonly referred to as UAT. This stage is different from the previous two in that it will not be focused on the technical side of the system. By the time UAT begins, the assumption will be that the features

are working correctly and that the system is stable enough to be used without constant supervision. What UAT is really concerned with is whether the system is actually practical and usable from the perspective of the people it was built for. A system can pass every technical test and still fall short if the people using it find it confusing, slow, or misaligned with how they actually work. UAT is the safeguard against that outcome.

For this phase, selected participants from D Speedway will be asked to use the system in an environment that closely mirrors their actual working conditions. The participants will include both the shop owner and staff members who are directly involved in the day-to-day processes the system is designed to support. They will not be asked to use the system in an artificial or simplified setting — they will be going through the same kinds of tasks they handle on a regular basis but doing those things through the new system instead of however they currently manage them. This real-conditions approach is important because it is the only way to find out whether the system holds up in practice, not just in a controlled test environment.

The following are the specific tasks that UAT participants will be asked to carry out during the evaluation sessions:

Logging in to the system using assigned credentials and navigating to the main dashboard — evaluating whether the login process is straightforward and whether the dashboard gives them the information they need without requiring additional steps.

Recording a sales transaction involving multiple parts — evaluating whether the sales form is easy to understand, whether part selection and quantity entry feel natural, and whether the confirmation after saving is clear enough that they feel confident the transaction was recorded correctly.

Checking current stock levels for specific parts using the inventory section — evaluating whether finding a specific item is quick and whether the stock status indicators are clear enough to understand at a glance without needing an explanation.

Responding to a low-stock alert by navigating to the reorder management section and creating a reorder record — evaluating whether the alert is noticeable enough to catch attention during a busy working day, whether the reorder form is simple to fill out, and whether the supplier information is surfaced in a way that reduces the effort of placing an order.

Generating a sales report for a specified time period — evaluating whether the report is generated quickly, whether the figures presented are easy to read and interpret, and whether the report gives the owner the information they actually need to understand business performance.

Updating a reorder record from pending to received and verifying that the inventory count has been restored — evaluating whether the status update process is intuitive and whether the inventory change is immediately visible after the update is saved.

Navigating freely through the system to explore any section they are curious about — evaluating overall ease of navigation, consistency of the interface across sections, and whether any part of the system feels unexpectedly complicated or unclear.

Throughout each session, participant behavior will be observed closely. Notes will be taken on where they hesitate, where they make mistakes, where they look confused, and where the process moves quickly and smoothly. After each task is completed, the participants will be asked to share their honest reactions — not just whether they managed to complete the task but whether they found it easy, whether it felt natural, and whether there was anything about the process that they thought could be better. These impressions will be captured in as much detail as possible because the kind of feedback that comes from actually using a system is different from the kind that comes from looking at it on paper.

At the end of the UAT sessions, each participant will be asked to complete an evaluation form based on the ISO 25010 quality standard criteria. This standard assesses software quality across several dimensions that are directly relevant to a system like this one. The evaluation will cover the following areas:

Functional Suitability — Whether the system does what it is supposed to do and whether the features available match the tasks the user actually needs to perform during a working day at D Speedway.

Performance Efficiency — Whether the system responds quickly enough to feel usable in a busy shop environment where staff do not have time to wait for screens to load or for operations to process.

Usability — Whether the interface is easy to understand, whether navigation between sections is clear, and whether someone with limited technical experience can use the system confidently after a short orientation.

Reliability — Whether the system consistently produces accurate results and whether it behaves the same way each time the same task is performed, without unexpected errors or inconsistencies.

Security — Whether the login and access level controls work as intended and whether the system protects business data from being accessed or modified by users who should not have that capability.

Maintainability — Whether the system is organized in a way that makes it straightforward to update or adjust features in the future as the shop's needs change over time.

3.7 Evaluation Criteria

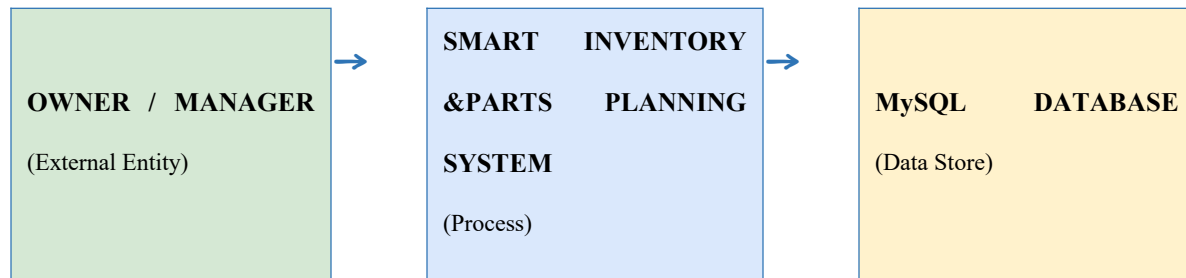
Once testing is complete, the system will be evaluated on the following:

- **Functionality** — Does every feature do what it is supposed to do?
- **Usability** — Can D Speedway's owner and manager actually use it without needing technical help?
- **Accuracy** — Are the stock counts, sales totals, and forecasting outputs correct?
- **Efficiency** — Does it take noticeably less time and effort to manage inventory and reporting compared to the current Excel-based process?
- **Reliability** — Does the system run consistently without errors, crashes, or data loss?

SUPPORTING DIAGRAMS

(Included within Chapter 3 — these diagrams illustrate the structure, data flow, and database design of the Smart Inventory and Parts Planning System for D Speedway.)

Figure 1: Data Flow Diagram (DFD – Level 0)



INPUT FLOWS	OUTPUT FLOWS
User records a sale or parts used	Inventory levels updated automatically
User requests a sales or inventory report	Report displayed on screen; CSV export available
System reads historical transactions	Forecasting results returned to dashboard
Stock drops below reorder threshold	Visual alert displayed on dashboard and inventory page
Manager manually creates reorder record	<i>Reorder saved with status: Pending</i> <i>→ Confirmed → Received</i>

Figure 1: Level 0 DFD — top-level data movement between users, the system, and the database.

Figure 2: Entity Relationship Diagram (ERD)

The ERD below reflects the actual database structure of the system, implemented in MySQL under the schema name login_system. The database contains ten tables covering every core function of the application.

users	products	categories
PK id : int(11)	PK product_id : int(11)	PK category_id : int(11)
name : varchar(100)	FK category_id : int(11)	category_name : varchar(100)
email : varchar(100)	code : varchar(50)	created_at : timestamp
password : varchar(255)	description : varchar(255)	
role : varchar(50)	unit : varchar(50)	
	unit_cost : decimal(10,2)	
	selling_price : decimal(10,2)	
	margin : decimal(10,2)	
	initial_quantity : int(11)	

reorder_threshold	:
int(11)	
created_at	:
timestamp	

product_stock	
PK	product_id : int(11)
code	: varchar(50)
description	: varchar(255)
unit	: varchar(50)
unit_cost	: decimal(10,2)
selling_price	: decimal(10,2)
margin	: decimal(10,2)
initial_quantity	: int(11)
reorder_threshold	: int(11)
FK	category_id : int(11)
category_name	: varchar(100)
current_stock	:

sales	
PK	id : int(11)
sale_date	: date
customer_name	: varchar(255)
plate_number	: varchar(50)
parts_total	: decimal(12,2)
labor_total	: decimal(12,2)
FK	created_by : int(11)
created_at	: timestamp

inventory_transactions	
PK	transaction_id : int(11)
FK	product_id : int(11)
transaction_date	: date
quantity_change	: int(11)
transaction_type	: enum
remarks	: varchar(255)
FK	created_by : int(11)
created_at	: timestamp

decimal(33,0)

sale_items	reorder_preparations	expenses	sms_settings
PK id : int(11)	PK id : int(11)	PK id : int(11)	PK id : int(11)
FK sale_id : int(11)	FK product_id : int(11)		
	recommended_qty :	expense_date : date	setting_key : varchar(50)
FK product_id : int(11)	confirmed_qty :	category : varchar(100)	setting_value : text
line_type : enum(parts/labor)	status : enum		
description : varchar(255)	(pending/confirmed/receive	description : varchar(255)	updated_at : timestamp
quantity : int(11)	confirmed_at :	amount : decimal(12,2)	FK updated_by : int(11)
unit_price : decimal(10,2)	created_at :	FK created_by : int(11)	
amount : decimal(12,2)		created_at : timestamp	
created_at : timestamp			

Table 1: ERD Relationships

ENTITY / TABLE	RELATED TO	RELATIONSHIP DESCRIPTION
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users	sales, inventory_transactions, expenses	A user (owner or manager) can record many sales, transactions, and expenses. All events are traced via created_by.
products	categories	Each product belongs to one category. A category can contain many products.
product_stock	products, inventory_transactions	product_stock is a computed view of current stock, combining product data with all inventory_transactions for that product.
sales	sale_items	Each sale has one or more line items (parts or labor) stored in sale_items.
sale_items	products	Each parts line item references a specific product from the catalog.
products	inventory_transactions	Every stock change (sale, restock, adjustment) is logged as a separate

		inventory_transaction record.
products	reorder_preparations	When stock falls at or below reorder_threshold, a reorder_preparations record is created. Quantity and restock details are entered manually.
reorder_preparations	(status flow)	<i>Records flow through three states: Pending (created) → Confirmed (manager approves) → Received (stock arrives).</i>

Figure 2: ERD — 10-table MySQL database schema. PK = Primary Key, FK = Foreign Key.

Figure 3: Use Case Diagram

The table below lists the main actors in the system and the actions each one can perform.

ACTOR	USE CASE (Action)	WHAT THIS MEANS
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Manager	Log In	Authenticate using email and password (bcrypt verified).
	Record a Sale	Enter sale date, customer name, plate number, parts used, and labor charges.
	View Current Stock	Check real-time quantity on hand for any part from the product_stock view.
	Receive Visual Low-Stock Alert	View color-coded badges on dashboard and inventory page when stock is at or below threshold.
	Restock a Part	Manually enter quantity received; system logs it as a restock in inventory_transactions.
Owner	All Manager actions	Owner has full access to everything the manager can do.
	View Sales & Inventory Reports	Generate daily, monthly, or yearly summaries from sales and inventory data.
	View Forecasting Output	See demand analysis per product based on inventory_transactions history.

	Manage Reorder Records	Create, confirm, and mark <code>reorder_preparations</code> as received. Quantity and supplier entered manually.
	Record Expenses	Log operational costs in the expenses table; reflected in dashboard KPIs.
	Manage User Accounts	Add, edit, or delete manager accounts. Owner account cannot be deleted.
	Export Data to CSV	Export sales history and inventory records for offline use or record-keeping.
System	Auto-update Inventory	After every sale, write an <code>inventory_transaction</code> record and recalculate <code>current_stock</code> in <code>product_stock</code> .
	Display Low-Stock Alert	<i>When <code>current_stock</code> $\leq$ <code>reorder_threshold</code>, flag item with yellow/red badge on dashboard and inventory</i>

		<i>page.</i>
	Run Demand Forecast	Analyze inventory_transactions history to display demand trends per product in the Forecasting tab.

Figure 3: Use Case Diagram — actions for Owner, Manager, and the system. SMS Notification is a planned feature not in the current build.

Note to the Instructor

CHED BSIT ALIGNMENT STATEMENT

This capstone project is aligned with CHED's BSIT competency standards. The work covers application development, relational database design and management, systems analysis and design, and the integration of emerging technologies including data-driven forecasting. The project was designed around a real local business, which grounds the academic work in an authentic context and requires the students to apply their technical knowledge to a genuine operational problem. Ethical considerations — including data privacy and responsible handling of business information — are incorporated into the system design. The project supports the outcomes-based objectives of the BSIT program and contributes to the professional readiness of the student-researchers.